

Down But Not Out

Canadian Forces Survival Manual (B-GA-217-001/PT-001)

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B-GA-217-001/PT-001

SURVIVAL
PSYCHOLOGY

CHAPTER 1

PSYCHOLOGICAL ASPECTS OF SURVIVAL

INTRODUCTION

In recent years many advances have been made in the development of clothing, equipment and rations for survival, and in the techniques for their use. However, regardless of how good equipment is or how good the techniques for its use are, the man faced with a survival situation still has to deal with himself. Man's psychological reactions to the stresses of survival often make him unable to utilize his available resources.

The information available in this field is far from complete but enough has been done to give an understanding of some of the major psychological factors involved in survival. While much of this information could be labelled common sense, it should be remembered that common sense is based on past experiences which have led to successful adjustments to various situations and that inability to use common sense under stress had led to the deaths of many very able, apparently sensible persons.

One of the most important psychological requirements for survival is the ability to accept immediately the reality of a new emergency and react appropriately to it.

Studies have demonstrated that survival information contributes to a feeling of confidence in one's ability to survive. Since self-confidence is important in handling fear and panic, survival information should serve to minimize fear and prevent panic.

Survival training should have in it some elements of realism like "battle inoculation" but safeguards must be taken against the production of casualties. Men must feel that they have come through real danger but have escaped unharmed.

Fear

Fear is a very normal reaction for any man faced with an emergency which threatens any of his important needs. Fear influences man's behaviour and thus his chances for survival; fear may ruin his chances or

may actually improve them. There is no advantage in avoiding fear by

(c) Thirst. Thirst is another enemy of survival. Even when thirst is denying the existence of danger. There is always something that can be not extreme, it can dull your mind. As with pain and cold, thirst done to improve the situation. Acceptance of fear as a natural reaction can be almost forgotten if the will to survive is strong enough. to a threatening situation will lead to purposive rather than random

It is also important to remember not to deprive oneself unnecessary of water. Serious dehydration may occur in a survival

How a person will react to fear depends more on himself than on situation even when there is plenty of water available.

the situation. It isn't always the physically strong or happy-go-lucky person

(d) Hunger. Hunger is dangerous because of the effects it can have

who most effectively handles fear; timid or anxious persons may respond

on the mind, primarily in lessening the person's ability for

more coolly and effectively under stress with a resulting better chance for

rational thought. Both thirst and hunger increase a person's

survival. Fear must be recognized, lived with, and if possible, utilized to

susceptibility to the weakening effects of cold, pain, and fear.

advantage.

(e) Fatigue. Even a very moderate amount of fatigue can materially

Factors increasing fear are mainly helplessness and hopelessness. Some

reduce mental ability. Fatigue can make you careless—it becomes

of the factors most frequently reported to decrease or help control fear

increasingly easy to adopt the feeling of just not caring. This is

are:

one of the biggest dangers in survival. The confused notion that fatigue and energy expenditure are directly related may be

(a) having confidence in your equipment; : responsible for many deaths in survival situations. Certainly,

(b) having confidence in the technical ability of your immediate

there is a real danger of over-exertion, but fatigue may actually

superior; and

be due to hopelessness, lack of a goal, dissatisfaction, frustration, an escape from a

or boredom. Fatigue may represent

situation

(c) concentrating on the job to be done.

which has become too difficult. If you recognize the dangers

a situation, you can often summon the strength to go on.

Seven Enemies of Survival

(f) Boredom and Loneliness. Boredom and loneliness are two of the

toughest enemies of survival. They are bad mainly

because they

Pain, cold, thirst, hunger, fatigue, boredom, and loneliness—everyone

are unexpected. When nothing happens, when something is ex-

has experienced these but few have known them to the extent that they

pected and doesn't come off, when you must stay still, quiet

have threatened survival. In the survival situation, the feelings of pain,

and alone, these feelings creep up on you.

cold, etc., are no different from those experienced elsewhere. They are

only more severe and more dangerous. With these feelings, as with fear,

the more you know about them and their effects on you, the better you

Attitudes for Survival

will be able to control them, rather than letting them control you.

While some attitudes are actually essential for survival there are others

(a) Pain. Pain is Nature's way of making you pay attention to some-

which greatly endanger chances of survival. For example, the mental

thing that is wrong with you. But Nature also has ways of hold-

attitude that "it can't happen to me" can blind you to the reality of the

ing off pain if you are too busy doing something else to pay

situation and make appropriate reaction in the face of emergency

impossible.

attention to the injury right then. Pain may go unnoticed if your

Much of the available evidence demonstrates the importance

mind is occupied with plans for survival. On the other hand,

ing a "preparatory attitude" for whatever emergency may

occur. This once given in to, pain will weaken the drive to survive. Pain

leads to preparation and rehearsal for all emergency possibilities. Survival

can get the best of you if you let it, even if it isn't serious or

instructions, given through posters, movies, lectures, etc, have been found

prolonged. A special effort must be made to keep hopes up and

to produce preparatory attitudes for survival which later serve as guides

to keep working.

to action. Either general or specific information may serve this purpose.

Often knowledge of the experience of others has also served as

preparation

(b) Cold. Cold is a much greater threat to survival than it sounds.

for action. A great number of incidents have been recorded which indicate

It not only lowers your ability to think, but also tends to lower

that previous rehearsal, both mental and actual, of emergency procedures

your will to do anything but get warm again. Cold is an insidious

serves as preparation resulting in automatic action.
enemy; at the same time that it numbs the mind and the body,
The most frequently encountered attitude endangering survival is the
it numbs the will. Because it is hard to move and you want to
"it can't happen to me" attitude. This is the opposite of the preparatory
sleep, you can forget your goal . . . to survive.
attitude.

Failure to have an attitude for survival may also result in panic,

Checking Equipment. In many more cases than will ever be even in persons who appear to be extremely calm under normal conditions. known, failure to check equipment resulted in failure to survive.

Knowledge and rehearsal of survival and emergency procedure not only bring about a feeling of confidence and preparation for survival, but

Surveying the Situation. The necessity for surveying the situation they can operate even when the survivor is in a state of semi-consciousness is widely recognized in problem solving. Frequently the things during an emergency.

most feared are those involving the unknown. Surveying the situation serves to remove those unknowns and to set in readiness powers of adjustment.

Group Behaviour in Survival

Survival Knowledge and Skills. Research has shown that con- Group Organization. A crew's chances of surviving depends largely on it's ability to organize activity. An emergency does not weld a crew fidence in one's ability to survive is increased by acquiring survival knowledge and skills. together; rather, the more difficult and disordered the situation, the greater are the disorganized crew's problems. This is particularly true in the face

Reaction Speed. In survival emergencies, previously established of common danger, when fear can result in panic rather than concentration. reaction patterns are important in enabling the group to react

Group Morale. High group morale exists when all crew members quickly. It is claimed that one main purpose of survival training is feel themselves part of the crew rather than individuals, and are proud to to afford an opportunity for knowing and understanding the be members of that crew. High group morale has many advantages.

responses persons and groups need to acquire. Well established

(a) The individual feels strengthened and protected since he realizes patterns of response may operate even when there is shock and that his survival depends on others whom he trusts.

panic.

(b) The group can meet failure with greater persistency.

(c) The group can formulate goals to help each other face the future.

Personality Requirements of Survival

High morale must come from internal cohesiveness and not merely Survival may depend more upon personality than upon the danger, nature of the emergency. Whether fear will lead to through external pressures. Under certain conditions, moods and attitudes terrain, or

become wildly contagious. Panic often may be prevented by conscious, act as a spur to greater sharpness, whether fatigue will overcome well-planned organization and leadership on the basis of delegated or shared person or leave him able to take the necessary action to survive, even responsibility, combined with faith in the group and realization of the or not he will have frost-bitten feet, all are, to a large extent, need for co-operation.

more on the person than on the situation. Qualities in a man

to survival are as follows.

Important Factors in Successful Group Survival
He can make up his mind.

(a) Organization of manpower. Organized action, when crew mem- He can improvise.

bers know what to do and when to do it, both under ordinary He can live with himself. Some people can't stand being alone, circumstances and in emergencies, is one good way of combating have to be entertained, etc. Others can take care of themselves, panic. An important technique for achieving organized action in make a good thing out of a bad one, keep their own and others' survival is to keep the crew well briefed. spirits up.

(b) Selective Use of Personnel. In well-organized groups, the person He can adapt to the situation. Some people can't change them- often does the job that most closely fits his personal qualifications. selves, no matter how much their stubbornness costs.

(c) Acceptance of Suggestions and Criticisms. Although some one He can keep cool, calm, and collected. person, such as the aircraft commander, must accept responsi- bility for final decisions, he still can make good use of others' He hopes for the best, but prepares for the worst. suggestions and criticisms.

He has patience. Some people must do everything right now.

(d) Consideration of Available Time. Hundreds of survival stories Others are able to wait until they have a surer chance.

indicate that there is rarely unlimited time in which to make He can take it. Few people know how much they can really take, decisions. On-the-spot decisions which must be acted upon im- but expecting things to be tough or unpleasant helps any of us

(e)

(f)

(g)

(h)

weather,

panic or

the

whether

dependen

important

(a)

(b)

(c)

(d)

(e)

(f)

(g)

a

(h)

mediately usually determine survival success,
to be prepared to meet the worst that can happen.

ee (j) He can figure out the other man. Some men step on everybody's toes, while others manage to keep practically everybody happy. The principal reason that this second group of men is so successful is that they are always aware of the feelings and moods

' of other people. This ability to understand other people and to

(k) predict what they are going to do is important to survival. He knows where his special fears and worries come from. Ail of us had accidents, scares, and worries when we were children that still bother us. Under dangerous survival conditions, cause trouble, but if a man knows where they came from, he can

SURVIVAL

these may

GEOGRAPHY

do something to control them,

CHAPTER 2

SURVIVAL

GEOGRAPHY

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INTRODUCTION

Geography in the survival sense must be somewhat restricted in scope. This is because 'a survivor will be interested only in the aspects of geography that directly pertain to him. The survival concept of geography may differ also from that of the botanist or the cartographer. For example, the Arctic is commonly considered to include everything north of the Arctic circle, i.e., everything above latitude 66°32' whereas the survival definition of the Arctic includes everything north of the tree-line.

The aim of survival geography is to teach the survival student what to expect from the land, wherever he may be. A survivor who knows what to expect will find that his problems are greatly reduced, for he should then know what to do. Once a survivor settles down to improving his situation, he will also improve his morale and strengthen his will to live.

Topography affects temperature, humidity, and watersheds which in turn influence the plant growth. The plant growth largely determines the animal life which can exist in any geographic region. The carnivorous predators, in their turn, depend on the seed and plant eaters and they consequently live a somewhat precarious existence which may be terminated by a change in any one of a number of factors.

The following map shows a simplified division of Canada's forested and non-forested regions. The break-down is made into eight parts, four of which are particularly important from a survival point of view. The remaining four parts are covered in order to fill in the background with additional information that may be useful. The emphasis is on northern survival because of the strategic and economic development of and the ever increasing amount of flying being done over Canada's northland.

It may be readily seen that the Boreal Region is the largest forest region in Canada. This region occupies a major part of the Canadian Shield, a massive formation of eroded palaeozoic rock, sparsely covered with a poor topsoil, which is unsuitable for agriculture but highly satisfactory for the growth of certain trees. A large portion of Canada's forest

wealth occupies the Canadian Shield and this great forest region shows remarkably little variation from one coast to the other. This is because the factors which influence forest growth and development, such as soil, temperature, humidity, and sunlight, remain relatively constant across this entire region. When one factor, e.g., temperature, does vary, it will be found that another factor, e.g., humidity, tends to compensate and the net result is the same or a similar type of growth.

Local variations will be found, however, and changes will occur, in the plant life within regions. The changes may be caused by such things as fire, disease, cutting, and even normal growth. In any event, the plant eating insects, birds, and animals will be affected by such variations, and the predator population will also vary according to the increase or decrease in the number of the various types of plant eaters.

Bearing in mind, then, that local variations are inevitable, the follow-

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ing generalized information which necessarily omits much detail, is offered as a guide to survival geography.

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a)

Se

Acadian (Maritimes and Newfoundland)

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Topography and soil--includes part of the Appalachians; picturesque,

S

rolling hills; wide river valleys.

&

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Watersheds--scattered small lakes well drained by extensive networks

2

of moderately flowing streams and rivers.

=

Climate and humidity -- averages 140 cm of rain a year; temperatures moderate; few extremes except in inland areas.

Plant life--spruce, balsam, yellow birch, maple, and pine; very

4

similar to the Boreal Region.

Wild life--trout, salmon, shad, alewife, eel, sucker, and chub; red squirrel, varying hare, porcupine, muskrat, beaver, weasel, mink, otter,

i)

we

fox, lynx, panther, black bear, deer, and moose; pheasant, partridge, grouse, duck, hawk, owl, eagle, gull, crow, and raven.

Great Lakes and St. Lawrence

Topography and soil--mostly peneplain; Canada's biggest fruit and vegetable growing district.

Rocky Mountain

....

rivers drain into the Great
Prairie GRASSLAND..

Watersheds--short, moderately flowing

TRANSITIONAL

.....

GREAT LAKES AND

Lakes and St. Lawrence.

St. LAWRENCE

Climate and humidity -- quite humid, mostly over 125 cm of rain a

West Coast ..

year; temperatures not very extreme.

Plant life--white and red pine, spruce, yellow birch, and maple.

Wild life--deer, grey and black squirrel, cotton-tail rabbit, jack rabbit, porcupine, fox, raccoon, and muskrat; hawk, owl, and pheasant; turtle, pike, smelt, perch, walleye, bass, catfish, and carp.

Prairie Grassland

in most lakes and streams; mice, red squirrel, varying
Topography and soil—the Great Plains are mostly very flat and have,
musk-rat, white tail deer, moose, weasel, mink, marten
except for broken up bad lands, in the south-west corner, extremely deep
wolverine, fox, wolf, lynx, and black bear; partridge, ruffed

soil.
owl, and eagle.

Watersheds—innumerable scattered sloughs and small lakes; slow
flowing rivers in deep coulees; often wide valleys.

Climate and humidity — dry climate, 25 cm of rain in the west to
soil—the same as the Boreal Region except for some
50 cm in the east; the temperature varies to extremes.
considerable muskeg.

Plant life—grasses, sedges, bullrushes, and bushes.
same as the Boreal Region.
Wild life—porcupine, gopher, prairie dog, rabbit, badger, coyote,
humidity—somewhat less precipitation than in the Boreal
deer, antelope, and elk; prairie chicken, hawk, owl, duck, and geese; pike,
with shorter summers.
jackfish, trout, and white-fish.
life—scrubby spruce and jack-pine.

Rocky Mountains
red squirrel, varying hare, fox, polecat family, and
Topography and soil—mountainous with some plateaus in the interior;
partridge, ptarmigan, and arctic owl.
rocky with thin sandy soil.

a
Watersheds—snow-fed lakes and streams; rapid run off in the spring;
swiftly flowing rivers.
soil

Climate and humidity — precipitation as high as 250 cm in the
Arctic—high rocky, shallow soil, permafrost, long fiords
higher areas; extreme variance in temperatures and humidity.
coast;
Plant life—Engelmann spruce, lodgepole pine, cedar, hemlock, larch,
and poplar.
barrens—low lying terrain;
Wild life—porcupine, hoary marmot, squirrel, rabbit, sheep, goat,
Arctic—low lying but rolling terrain; and
moose, deer, panther, and grizzly and black bear; trout, and no birds.

Islands—eastern islands mountainous, western islands flat.
West Coast
Topography and soil—wide fertile river valleys surrounded by
mountainous topography.
Arctic—glacier-fed rivers which dwindle in summer;
Watersheds—well drained by rivers flowing into the Pacific which
barrens—extensive muskeg, innumerable shallow lakes,
tend to flood during the spring run off.
shallow rivers;
Climate and humidity—very humid; high rainfall; mild climate;
maritime,
Arctic—few lakes or rivers except north of the main-
barrens.

Plant life—Douglas fir, western hemlock, western red cedar, and
Sitka spruce.
"humidity—long winters of darkness, surprisingly little
Wild life—porcupine, marmot, rabbit, grouse, mule deer, red deer,
cold; sea frozen from mid October to June; short
and bear; trout, salmon, various cod, and flounder.
daylight, seldom hotter than 10°C and lasting

Boreal Region
early August.
Topography and soil—lies on the Canadian Shield; shallow, poor soil;
life—scrub willow in stream beds and protected places; mosses,
overlying rock; rolling terrain; hummocky.
Watersheds—extensive muskeg; innumerable lakes; fairly fast flowing
life—abundant in fish; lemming, ground squirrel, arctic hare,
streams; most drainage to the north.

Wild life—fish

hare, porcupine,
(fisher), otter,

ge
grouse, duck, hawk,

Ae
Northern Transitiona

SER
Topography and
permafrost;

Log
Watersheds—the

Climate and
Region and colder,

Plant

Wild life—fish,

wolf; spruce

Arctic
Topography and

ee

(a) Eastern
by the

(b) Mainland

(c) Western

(d) Arctic

Watersheds
(a) Eastern

(b) Mainland

slow

(c) Western
land

Climate and

snowfall, extreme dr

summers of complete

through July and

Plant

lichens, and berries
Wild

fox, wolf, polar

bear, seal, walrus, whale, caribou and musk-ox; duck, geese,
Climate and humidity ~ precipitation variable 25 cm in the west to ptarmigan, and sea
birds on the coast.
25 cm in the east; cold winters; short hot summers.
Plant life—spruce, balsam, white birch, poplar, jack-pine, and
tamarack.

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FIRST AID

CHAPTER 4
FIRST AID

GENERAL

Definition

First aid is immediate assistance rendered to the injured or suddenly ill person in the absence of adequate medical care. ;

Purpose

To prolong life, to preserve existence and vitality, and to prevent further injury.

General Rules

- Check bleeding and breathing.
- Do not get excited but act quickly after you decide which of the injuries needs attention first.
- Do not move the patient unless he is certain to benefit as a result.
- Keep the patient comfortable and warm; reassure him and avoid having him see his injuries if possible.
- Do not touch open wounds or burns with your fingers or other objects.
- Do not give unconscious patients liquids.

SHOCK

Definition

Shock is a condition which is caused by the loss of an effective volume of blood circulating in the patient's blood vessels. It may be caused by different factors, such as:

- (a) actual loss of blood from: wounds;
- (b) loss into the tissues such as in a broken thigh – as much as 2.2 l may pool in the fractured leg;
- (c) internal bleeding into body cavities;
- (d) loss of other fluids such as serum and plasma in burns; and
- (e) massive loss of fluid through sweating, vomiting, and diarrhoea.

Signs of Shock

The skin is pale, cold, and clammy, the pulse is fast and weak, and the breathing is fast and shallow.

Treatment of Shock

is coughing blood as from a damaged lung, sit
Control bleeding:
his head supported; and
(a) external signs
is vomiting blood as from a damaged stomach,
(i) arterial—bright red—spurting,
knees, make him as comfortable as possible, and ensure
(ii) venous—dark red—rapid flowing, and
passage is clear.
(iii) capillary—steady oozing; and
(b) internal signs
bodies from the surface;
(i) luags—chest pains—bright frothy blood from the mouth, and
the wound;
(ii) abdomen—stomach injury—vomiting blood of coffee ground
sterile dressing;
colour.
fractures; and
Stop bleeding by a pressure bandage at the site. Complete rest is essential.
compresses to combat infection.
Facilitate or restore breathing:
(a) by artificial respiration; or
(b) by giving oxygen if the aircraft supply is available.
the wound air-tight.
Control pain. Severe pain must be relieved. Pain can often be eased
by keeping an injured person quiet and warm, carefully changing his
patient in a comfortable position;
position to keep him comfortable, splinting an injured arm or leg, and
by mouth; and
handling him gently at all times. Give 222s in accordance with instructions
binder type dressing.
when a severely injured person, suffering acute pain, has to be transferred
to a safer location, or when severe pain cannot be controlled by any
FRACTURES
other means. Never give them to a patient who is unconscious or sleeping,
who has @ severe head injury, or whose rate of breathing is twelve respira-
indications of fracture:
tions or less a minute.
snap;
Prevent additional injury:
site;
(a) protect the injury;
of the affected limb;
(b) splint any fractures before moving; and
(c) handle gently.
Replace fluids. Give hot sweetened drinks,
and
Prevent exposure:
(a) protect under the patient as well as on top; and
(b) do not overheat.
sedatives;
The head should be lower than the feet.
fracture and neighboring joints; and
Reassure the Patient
contamination.

WOUNDS

Preliminary Treatment

Spine:
Expose to determine the extent of the injury,
Stop bleeding:
Pain, tenderness, or deformity at the site of the
possible paralysis or loss of sensation below the site
(a) apply a firm pressure dressing and in extreme bleeding cases
apply a tourniquet (the tourniquet should be used only as a last
resort);
Fractures of the spine should be treated with great
of the danger of injury to the spinal cord. If the

(b) if the patient

him up with

(c) if the patient

bend his

that his air

Protect the wound:

(a) remove foreign

(b) do not probe

(c) apply a

(d) dress and splint

(e) apply hot steril

Complicated Wounds
Chest wounds. Make

Abdominal wounds:

(a) place the

(b) give nothing

(c) apply a large

Indications of Fracture
The following are

(a) an audible

(b) pain at the

(c) loss of power

(d) deformity;

(e) tenderness;

(f) swelling;

(g) discoloration

(h) crepitus.

General treatment:

(a) administer

(b) immobilize the

(c) avoid

Special Fractures

Fracture of the

(a) Indications.

fracture and

of the fracture

(b) Treatment.

care because

back is

allowed to hump, the broken bone may cut the spinal

cord.

Fractured Ribs

(a) Indications. Pain, especially on breathing or coughing. The broken ASPHYXIA

rib is tender and the break can sometimes be felt. The patient unconsciousness from lack of oxygen.

usually holds his hand tightly over the break. If the lung is punctured he may cough up bright red frothy blood.

or the patient from the cause (protect yourself);

(b) Treatment. Tie a firm bandage round the lower ribs. passages, loosen clothing, and remove false teeth;

Fracture of the Skull

artificial respiration and give oxygen if available;

(depressants) or alcohol; and

(a) Indications. Unconsciousness, swelling, or laceration of the

scalp; bleeding or leakage of spinal fluid from the nose, mouth, or ears; a difference in the size of the pupils of the eyes, blackening

HEAT STROKE – SUN STROKE

of the tissues under the eyes, and paralysis or twitching of the regulating mechanism of the body fails and the body muscles.

dangerously high levels.

(b) Treatment. Treat as for shock. Keep the patient at rest in a semi-prone position.

and dry;

Fracture of the Lower Jaw.

a

(a) Indications. There may be pain on movement of the jaw; irregularity of the teeth, inability to talk or swallow in some cases, and bleeding from the mouth.

disturbances; and

(b) Treatment. Clear the nose and keep the mouth closed. Immobilize vomiting.

using bandages,

Fracture of the Collar Bone.

quickly, particularly about the head area;

solutions by mouth; and

(a) Indications. The injured shoulder is at a lower level than the stimulants.

uninjured one. The patient is unable to lift his arm above his shoulder, and he supports the elbow of the involved side with the

FROST BITE

opposite hand. The fractured ends can usually be felt under the symptoms: skin.

dull whitish pallor;

(b) Treatment. Use a T splint or figure eight bandage with the centre of numbness or prickling associated with the of the eight between the shoulder blades, and small arm sling to crystals in the tissues;

support the shoulder on the injured side.

the tissues are solid and immovable; and

Dislocations, Strains, and Sprains. Damage to cold causes the person to become numb

to ligaments, tendons, and

muscles. Treat as for fractures, eyesight fails, he becomes unconscious, and his

cease.

BURNS

\

rab.

Burns are extremely painful and the resultant shock is more severe than

part is on the face, ears, or trunk, cover it with that encountered in most other injuries.

hand. If a hand is involved, insert it within:

Treatment:

against the body. If a foot is involved, remove the

(a) treat for shock;

place the foot within the shirt and against the

(b) give 222s;

man.

(c) avoid contamination of the burned surface; wrap in sterile dressing, cover warmly, and put

This is

Treatment:

(a) remove the cause

(b) clear air

(c) administer

(d) give no sedatives

(e) treat for shock.

The temperature

temperature rises to

Signs and symptoms of

(a) skin flushed, hot

(b) headache;

(c) dizziness;

(d) irritability;

(e) visual

(f) nausea and

Treatment:

(a) cool the patient

(b) give salt

(c) give no

Signs and

(a) the skin assumes

(b) there is a feeling

formation of ice

(c) with deep freezing

(d) prolonged exposure

and drowsy, his

respiration may

Treatment

(a) Do not

(b) If the frozen

the warm ungloved

the shirt, up

shoe and sock and

body of another

(c) Treat as a burn,

(d) cover the injury with sterile vaseline impregnated gauze;
rest.

(e) over this place a thick layer of sterile gauze dressing or similar
patient from exposure and give warm drinks, food,
sterile padding; and

Artificial respiration, stimulants, and oxygen may

(f) bandage firmly.

cases of prolonged exposure.

at complete

(d) Remove the
and clothing.
be necessary i

	TRIANGLE	BANDAGE	FOREIGN	BODIES
Ears	To dislodge lukewarm water.	a foreign body Be sure to	from direct	the ear, syringe the flow of the water along the side of the canal. If the object does not come out, do not try to dislodge with
L\FS				
pins, wire, etc.				
Nose				
	This usually presents no immediate danger. The object can generally be dislodged by stopping the unaffected nostril and blowing the nose. Any attempt to dig the object out will cause more swelling and lodge the object more securely,			
Eyes	Do not attempt rub the eye,	to remove	the foreign	object with' the fingers or to
	a			
Triangular Bandage	Folded	as a Cravat		
(a)	Close the eye for a few minutes until the spasm of irritation is over, then grasp the lashes of the upper lid, and raise the lid. Repeat this process a few times. In many cases the object will be washed out by tears.			
(b)	When this method fails, a search must be made under the lower lid. Place the thumb below the eye and pull down. This exposes the under surface of the lower lid for examination. The foreign body if seen can be picked off with a cotton tipped stick. If the foreign body is not seen, examine the upper lid. Have the patient look down, place a match across the upper lid, and then with the other hand grasp and raise the lashes, turning the lid back over the match. Lift out the foreign body with a cotton tipped stick. Place some eye ointment on the inner surface of the lower lid.			
(c)	If the object is embedded in the eye or lid or if there is difficulty removing it, close the eye and apply an eye patch held in place by strips of adhesive. Do not use a knife, toothpick, pin, or any similar object to remove foreign bodies from the eye.			
		INSECT	BITES	
Mosquitoes, hornets sting.	black flies, deer flies, and		midges bite. Bees, wasps, and	
Treatment				
(a)	Prevention is the best cure for insect bites. Keep yourself protected at all times with head nets, gloves, light coloured clothing, fly repellents, etc.			
(b)	Soaking in warm water is the easiest and best all around treatment for bites.			
(c)	A paste of clay and water will reduce the sting of the pests.			
Triangular Bandage	Used	as a Sling		

	Hand Bandage	.	Knee Bandage		
			Doughnut Bandage		
	Foot Bandage		or	Fracture of the Humerus	Immobilized by Binding to the Chest
Wall			Ring Pad		
		42			43

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THE ROLLER BANDAGE

Starting the Roller Bandage
for Joints Spiral Reverse Bandage Figure Eight Bandage
(Hold the roll in the right hand, loose end on the bottom.) {For tapered
surfaces—overlap each turn one third.)

Anchoring the Roller Bandage
(Make a few turns in the same spot.)
Start of the Recurrent Bandage

Fastening the Roller or Spiral Simple Spiral Bandage
Bandage (Used in bandaging cylindrical parts; Padding a Wooden
Splint Applying a Splint
(Tear the end into two tails, tie a overlap each turn one third.)
knot, carry one tail in the opposite
direction, and tie.)

SPLINT FOR FOREARM

Sticks Rolled in Cloth to Form an Improvised Splint for the Forearm.
Splint for Fracture of the Clavicle.

Immobilizing the Clavicle. Use two belts in figure eight fashion.

Sticks Rolled in Blankets to Form an Improvised Splint for the Leg.
placed over clavicles and under arm-pits beneath belts.

Method of

Pads should be

B-GA-217-001/PT-00!

SHELTERS

CuarTer 5

SHELTERS

INTRODUCTION

Once first aid has been administered to the injured it becomes necessary to provide a comfortable form of shelter. There are many types of shelter that are quickly and easily constructed and the selection of the one to build depends on circumstances, such as the availability of materials, season, geographic location, topography, etc. A few of the recommended types of shelters are:

- (a) lean-to;
- (b) suspension tepee;
- (c) cabin or para-cabin;
- (d) natural shelter;
- (e) aircraft; and
- (f) arctic shelters.

The Lean-to

A pole framework is covered with a thatching of parachute silk, evergreen boughs, rushes, heavy grasses, slabs of bark, or split wood planks. When constructing the lean-to, find two trees two to three metres apart with fairly level, firm ground between them. The distance between the trees will be the length of the opening of the lean-to, although it is possible to incorporate variations. The number of people requiring shelter should determine the size. When constructed for one man it should be made long enough for him to sleep across the open mouth of the shelter, whereas for more than one it should be planned for them to sleep lengthwise. One or both ends of the ridge pole may be supported by a pole bipod or tripod instead of utilizing standing trees. This leaves the builder a wider choice of sites. It should be remembered that the steeper the slope angle of the roof the better it will shed precipitation and reflect heat from the fire. A 45 degree slope angle is generally considered a suitable compromise between available interior space and rain shedding effectiveness.

Si

covering.
 Once the framework has been constructed, proceed with the although the branches
 lent natural covering,
 Spruce boughs make an excellent natural covering, d
 of any coniferous and of many deciduous trees will do. They are placed
 as shingles on a roof, the first row at
 on the lean-to in the same manner overlapping
 re the bottom. The brush ends of the boughs are placed down, of thatching ensures that
 the butt ends of the previous row. This method of boughs in this
 rain will be shed more readily. Continue to lay rows to is covered. Then repeat
 fashion until the top or back roof of the lean- to a depth of at
 | the thatching procedure until the entire roof is covered d in with large boughs set butt
 sides are filled
 least 15 cm. The triangular
 end up as in thatching.

Lean-to

The lean-to is an excellent shelter even in winter
 will since
 utilization of any desired type of fire. With a low mouth, it permits
 greater than the width of a sleeping bag, and the full length of the body
 See the fire, a person can be surprisingly comfortable in the coldest
 weather. It is possible to build a shelter of this type to house several
 sap but it is not nearly as comfortable sleeping with either head or
 om to the fire as it is sleeping crossways. Two lean-tos may be constructed
 facing each other to share a common fire but it is difficult to arrange th
 ars to avoid drawing smoke back into one or the other of them. You

succeed at first only to find that a slight wind sets up eddies
 | ies whi
 | mae he smoke unbearable. Two lean-tos facing each other and bisa
 | cs vce eas ean
 | lop will make a pup tent.

me

The Suspension Teepee
 lines of your to covering and
 Parachute silk or wing covers provide excellent Jean-
 This is a simple quickly erected teepee. Cut the shroud
 5 uncti on with ever green boughs. If
 or in conj
 parachute about half a metre from th
 \ can be used either in lieu of
 the temperature is very low, place the fabri
 c on the framework first, and
 shroud line around the cords in the mee ae Lamia sie eis of
 cloth on the inside of
 tree, a crossbar and throw the:
 the thatch with boughs on top. The light colored
 !
 Tope over an extended limb of a of at least Be rier frees or
 fire and provide greater warm
 the shelter will reflect the heat from snow, the, put
 a apex of a tripod made from poles etres
 the boughs on the framework
 isi done, secure the canopy at the desired height and ape
 If there are heavy rains or melting If
 ci
 will aid in keeping the shelter dry.

periphery with pegs, rocks, or sod, >
 thus forming a conical tent. The ie Cir-
 / first and the cloth on the top. This
 | available, a double layer of fabric is advisable.

't \
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cumference of the tent will be determined by the number of inhabitants and in most cases several gores of silk will be surplus. If there is a need for additional silk for other purposes, these extra gores may be cut out, or they may be left loose to form the entrance.

If it is planned to build a fire in the shelter, it is necessary to provide an adequate vent before elevating the canopy. In wooded areas, where winds are generally not strong, the hole in the apex is too small for proper ventilation and it is necessary to slit the canopy a few centimetres down along the centre channel of a gore seam. Keep the vent hole open by tying in a cross of sticks the desired opening size. Build the fire in the centre of the tepee, directly under the vent hole.

Suspension Tepee

that used by the Plains Indians, is

Another type of tepee, similar to the one shown below.

Smoke-filled Tepee

When mosquitoes, black flies, and other insects are numerous, a smoke-filled tepee may prove to be less objectionable than the blood-thirsty myriads outside.
Tree

Centre

Cabin or Para-cabin

This shelter requires a considerable amount of work and when completed will have a degree of permanency that the previously mentioned shelters do not have. When building a cabin, particular attention should be paid to the choice of location, since the cabin will not be portable. Choose an area close to a water supply, yet not in a valley. The ridges offer much more comfortable living conditions, freedom from insects, and so on. The area chosen should also offer an abundant supply of long straight logs, 10 to 20 cm in diameter. The simplest method of construction is to build four walls log cabin fashion to a height of about one metre and then build a framework of light poles to support a covering of parachute material.

of parachute material it is necessary
height and to add a sod or thatched
build an elaborate gable roof. A flat
that is required. The lower the roof,
some form of stove for cooking and
be sufficient for an open fire under

particularly in hilly or mountainous areas
and seas. Caves should not be over-
must remember that caves are fre-
already occupied. The base of a large over-
sufficient protection from the elements.

tree, if lying in a secure position, can be

From this stage it is a simple matter to take a parachute and place
by laying boughs or some other

roof,
it over the framework with the apex of the chute at the top of the
branches. Care should be taken that
preferable to employ a double layer of
fallen tree, which may be keeping it
to form a finished shelter. It is

insulating
fabric with an air space between, when possible, to improve the
base of a large tree with thick low
and water shedding qualities of the roof.

To build a cabin without the use
to construct the walls to the desired
roof, Do not make any attempt to
roof with a slope to the rear is all
the easier it is to heat the cabin.
This type of shelter requires
heating as the ventilation will not
formal conditions.

Natural Shelters

Caves may be encountered,
and by the shores of rivers, lakes,
looked as potential shelters but one
quently damp and are sometimes
hanging rock will sometimes provide

A reasonably large, fallen
cleared out underneath and closed in

siding material over the protruding
branches on the underside of the

off the ground, are not removed. The
branches can also be used.

Aircraft

The Fighter Trench

In warm weather, the aircraft fuselage, if sufficiently intact, and portions of the wing and tail can be made into a comfortable shelter. The does not permit building an igloo, the survivor should for main consideration will be the danger of spilled gas and fumes. The aircraft fuselage does not make a good shelter during winter because the the result achieved. metal will conduct what little heat is generated away from the shelter. If drift of snow at least one metre deep is available, the emergency tents can be salvaged from the wreck, then the survivor needs made by cutting large vertical blocks from a trench just only to pick a good location. sleeping bag and long enough to accommodate 'the one or

If time the first night minimum work for If a large shelter can be wider than the two builders. The

snow blocks are stood on each side of the trench.

Arctic Shelters

Tools. The combination snow saw-knife in your survival kit or a snow knife is essential to survival north of the tree line. With it you can cut snow blocks to build yourself a shelter.

Snow Saw-Knife

Cutting the Snow Blocks

Material. The snow from which the snow house is built is in a firmly packed and frozen form which has several characteristics not often trench is completed a notch is cut along each side to encountered south of the tree line. It is solid enough that a cubic foot support for the snow block roof. block will support the weight of a man, yet it can be cut, sawed, or split with ease. Even in the Arctic only a small percentage of the snow is suitable for snow house building. First look around for an area where snow-drifts are deep enough to permit cutting snow blocks from a vertical face. This will require a depth of nearly 60 cm. The snow should be firm enough to support your weight with only slight marking by footprints. Probe into the snow with your saw-knife or a long half a cm rod. Try to find a place where the resistance to the probe indicates an even firm structure, free of harder or softer layers. When you find a spot, probe around to ascertain whether enough good snow is available. It is well worth hunting for an hour to find proper snow as you will save the time during snow house building. If snow of sufficient depth to cut vertical blocks cannot be found it will be necessary to cut them from the flat surface of the snow. This is time consuming and requires a much larger The End Block The Trench area of snow, and the snow house will have to be built higher, because it 'cannot be dug into the drift.

When the provide non-slip

block is placed at one end of the trench as a support block of the roof.

A triangular for the first snow

The first roofing snow block is out shorter than the others, in order that the succeeding blocks will overlap, each supporting the next. be found, a trench style shelter can be

blocks enclosing the shelter area. This wall

slabs which are hollowed slightly on the

arch.

Commencing the Roof

The remainder of the roof blocks are placed in a similar manner. In a two man trench an entrance door is placed halfway down the trench, opening into a roofed over square pit which allows room for cooking and removing clothing before entering the sleeping bag. Be sure to cut a ventilating hole in the roof and have a good snow block handy to close the entrance at night. If snow depth permits, a pit or "cold well" should be dug out just inside the entrance to provide a lower heat level. This will also ease the problem of dressing and undressing.

Trench Style Shelter

good emergency shelter is too cramped to dislodging the frost on your clothing and will become damp without a good means of should begin your igloo as soon as you can.

Eskimo origin, and in that language it is a "shelter". In this manual it will be used to similar to that used by some Eskimo groups Arctic.

ideal winter shelter in the Arctic. It is solid, and it is large enough for comfort. There which must be mastered but no one of

Once the method is learned, the igloo will be used in an emergency.

If no deep snow drifts can erected by building a wall of is then roofed over with large inside, after erection, to form a

The fighter trench while a permit much movement without sleeping bag, and in time you drying out. This is why you

The Igloo

The word "igloo" is of general word for "house" or mean the domed snow house, particularly in the central

The Eskimo igloo is the sound-proof, and wind resistant, " are a few building techniques these is particularly difficult. almost invariably be the shelter

When you have found a good snow-drift, lay out the floor plan. The Eskimo does this by eye, but he has had a lot of practice. Draw a circle centred on the best snow, with the approximate diameter as follows:

One	man	—	2.5m
Two	man	—	27m
Three	man	—	3m
Four	man	—	3.7m
Five	man	—	4m

Now, begin to lay in a supply of snow blocks. Cut them from the face of a trench, laid out as shown.

Shaping the Block

block.

Now undercut the

Begin cutting blocks by digging out a clear vertical face at A-B, with a width of about 117 cm and a depth of about 50 cm. Smaller

Undercutting the block

blocks are not much easier to cut, and igloo construction is slower and more difficult with them.

With your snow saw-knife, cut a slot at each end of the block, ° Next, score a groove parallel to the face, marking off a block about 15 cm thick.

15 cm thick.

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B-GA-217-001/PT-001

reaches the snow block trench, a snow block is
to be taken across it. ,

When the first row
replaced in it to permit the wal

Cutting Out the Block

First Row Completed

The

Commencing to Build

q | If you have a snow saw-knife saw along this mark, breaking off the
block with a firm jab in the centre. If you are using a snow knife, deepen
the groove by running the point back and forth, then three or four gentle
Stabs and a firm central stroke will break it off.
Lift the snow block to one side and begin another. When you have
about a dozen cut, then you may begin to build.

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Note the slope of the first row of blocks. All end joints are fitted with faces radial to the igloo centre.

aa

See

Carving the Spiral

Correct Slope

Va

1

Nae, Incorrect Slope

Commencing the Second Row
Don't make this mistake, or you are heading for trouble.

the

"

When the first row is finished, begin the spiral which will end at the slope the other way.

If you are a southpaw, cu

key block. If you are right handed, cut away any three blocks diagonally ,

Now fit the next black,

leaning it inward so that its inner face is sloping down from left to right.

roughly tangential to the dome

66

67

Even at this early stage, the block might fall in, except that it is supported by the face of the notch and the top of the previous blocks.

Pressure
area
VY

te
Sa

Inward Pressure

Bearing Surfaces of the Block

inward around A-B is resisted by pressure
the faces of the new block and the previous
radial to the igloo centre, or the previous block

blocks on the A-B-C method described, the block
position and the joint fitted roughly, with the faces
supported by the left hand. If the snow saw-knife
and the previous one and the kerf pushed

undercut on the under face at the end nearest the
The block must bear only at areas A, B, and C. It should not bear at D

joint supported at A and C only.
or E or it will pivot and slip. All blocks from this point on, until the key
block, are set in this manner.

Continue cutting blocks from within the igloo circle, fitting them as
you go. Don't use blocks less than one metre long or half a metre wide

if you can help it. Lay small blocks aside for later use in snow bench
and doorway building. The slope of the block, which of course governs
the shape of the igloo, is estimated by eye. The block is raised into place
and the joints are trimmed until the block settles into position.

When the third row is under construction, the slope will be great
enough to make careful fitting essential. Each block bears in the same
three positions only. The remainder of the joint can gape wide, or almost
touch, but these three faces must carry the load to jam in the block.

Pressure Bearing Surfaces

The tendency to rotate
between the upper third of
block. This face must be
may be displaced.
When fitting snow
should be lifted into
in contact and the block
is run between the new block
closed, then a slight
ag a eg
previous block will leave th

ay

Inward

A firm tap downward at C as shown by the arrow will drive the block into final position, seating at A, B, and C, when it need no longer be supported.

Carry on building, block by block. You will find that the increasing slope of the igloo wall will of course increase the tendency for the block to fall in, but this is compensated by the increasing angle between the A-B axis of the successive blocks as the diameter of the opening decreases. Building actually becomes easier toward the finish, as the blocks will jam firmly into place.

When you run out of snow block snow inside the igloo, cautiously cut a small door as far down the wall as you can, tunnelling underneath to make enough space for the outside workers to push in more building blocks.

Try to keep the curve of the walls symmetrical and avoid a pointed igloo, because the high ceiling would reach the iimiting warmth before the sleeping bench gets its share of heat.

<sml
C109 YB

Fitting the Key Block

= -159C

ANI

*

\

Correct

should be longer than it is wide, to permit passing the

through, then juggling it into position. This is tricky, but

It is surprising how flat an arch can be built, using the spiral technique. to fail. By judicious use of your snow knife, cut away

The last few blocks will be almost horizontal, but if you remember the it settle slowly into position. You have built your igloo!

A-B-C fit, they won't fail.

When the remaining hole in the roof is small enough to permit doing Habitable

so, a key block is fitted. After what you have been doing, this is easy.

The edges of the hole should be bevelled at about 15 degrees from the floor, about one third of the way back from the door,

tens

vertical.
about 50 cm high to conserve warmth.

The hole

key block up

no one ever seems

the block, letting

Making the Igloo

Across the

build a snow wali

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N\A
Sleeping
shelf ¥

nen

\ _

Plan View of Igloo

heating area. It must be reasonably close to the

Iamp tender to reach it without rising from

Cross-section of Sleeping Shelf

igloo carefully with powder snow, which when

This will form the front of your sleeping shelf, which will raise you

seams will soon harden and stop loss of warm

into the warm air trapped above the door.

plan a short stay, chink only the outer seams,

Shove all the loose snow in the igloo behind the wall to form the shelf.

inside and outside joints.

Break up lumps and blocks to soften the bench and to provide better

3

insulation. Level the bench top carefully.

powdery snow on top of the igloo to act as

At each side of the door leave or erect little benches allowing about

to add to the weight of the Toot _

50 cm of leg room between the sleeping shelf and bench.

bottom row of blocks to prevent wind driven snow

This is the kitchen and

bench to permit the cook and

the sleeping bench.

Chink the dome of the

packed firmly into the open

air from the igloo. If you

but for a better job do both

You may throw loose

chinking, but net so much as

. You may bank the

} from causing erosion.

All snow and frost must be removed from hides, bedding, and clothing before they are placed on the sleeping bench.

kovik
y SES ventilating
s << gp hole
L) banking
E | 1 block il
Py high FON
ND, 5 bench
G)| 6
B R
714 B
A O N\ "Se an \, [loec change
s WN
Igloo Cross-section
comp primus
utensils stove

A Suggested Floor Plan of the Igloo

Pots can be suspended from pegs driven firmly into the walls above
It a high wind is blowing, the drifting snow can' erode the wall of
the fat lamp (koodlik) or the primus stove.
the igloo very rapidly. A snow wall should be erected to act as a
Drying racks made by forcing sticks into the walls above the heat
wind-break, and any broken blocks can be piled against the windward
sources will serve the following purposes.
wall to protect it from the cutting effect of the drift.

E 7 , +
Now, with the igloo chinked, the door cut in and the sleeping bench
(a) Drying of clothing from which all snow, ice, and frost have first
completed, all you need to do before moving in is to clear out all loose
been scraped. Never melt snow on garments—always scrape it off.

| snow. The bench is first covered with caribou skins (or other insulation)
(b) Thawing of frozen rations which do not need cooking. This
and the sleeping bags are then unrolled and placed, heads to the entrance,
— Requires quite a long time.
side by side.
(c) Protection of the igloo wall and roof from melting.

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cluttered with odds and ends not needed
wall,
be built against the outside
low to
cache. Keep the entrance
If
the igloo will warm up rapidly.
ice, and drip, you are overheating. Take
develops; cut down the heat if you must.
have no place in igloo living. Boiling and
be
satisfactory. Canned goods may
pot of water
unopened to a boil in a
covered
pressure cooker or a tightly
can over direct heat! This makes a fine
a fool rarely makes this mistake twice!
breakfast and the main meal in the evening,
activity to
at noon will not bring
so. Body heat is derived from food intake,
supplement with fish whenever possible. Eat fats
for
low. A diet of meat is good
superstition to the contrary. Vilhjalmur
point. If
meat alone to prove this
must eat
products of the chase, you
Persons entering the igloo for a stay of longer than an hour or so,
part to ensure that you don't suffer from
after removing mukluks and snow from garments, should get up on the
use as a
sleeping bench, out of the way.
on the floor for
The cook, usually at the right-hand bench, has the primus stove,
door is closed for the night. The user is
3
under which is a piece of cardboard from a ration box to prevent it from
disposition.
melting into the shelf and tipping. He may also have a koodlik, if fat is
available, for slow cooking.

One man should be responsible for adequate ventilation—keeping the
vent holes in the dome and door open enough to avoid risk without
freezing the occupants. Carbon monoxide is insidious and dangerous.
During the day the door is left open. At night it is closed by a snow
A avs Mary ac it 7
block which should be chinked and a ventilation hole should be bored
through the upper part. The more fumes being generated, the larger must
be the aperture. Don't wait until the lamp won't bum properly and you
begin to feel groggy before letting in air. It is dangerous, and it isn't
necessary at all. If the roof hole does not draw properly because of wind,
a snow chimney can be made by setting a perforated block over the hole.

If the group finds the igloo
at the moment, a miniature igloo can
and a doorway cut through to form a
avoid loss of heat.
Now that you are in residence,
the inner walls start to glaze, for
corrective action before icing
Frying, baking, or broiling
stewing are easier and prove very
heated in the can by bringing them
which completely covers them. Use the
pot to avoid steam.
Never place an unopened
bomb or grenade, and even
Two good meals a day,
avoid loss of the working day. A snack
a halt for more than an hour
eat all your ration and
so
rather than burn them if the supply is
despite some stubborn
you,
Stefansson lived for a full year on
you are forced to live solely on the
flesh, fat, liver, and every edible
dietetic deficiencies.

A snow block (Kovik) may be kept
out after the snow block
chamber-p
responsible for its sanitary

SHEE

AW

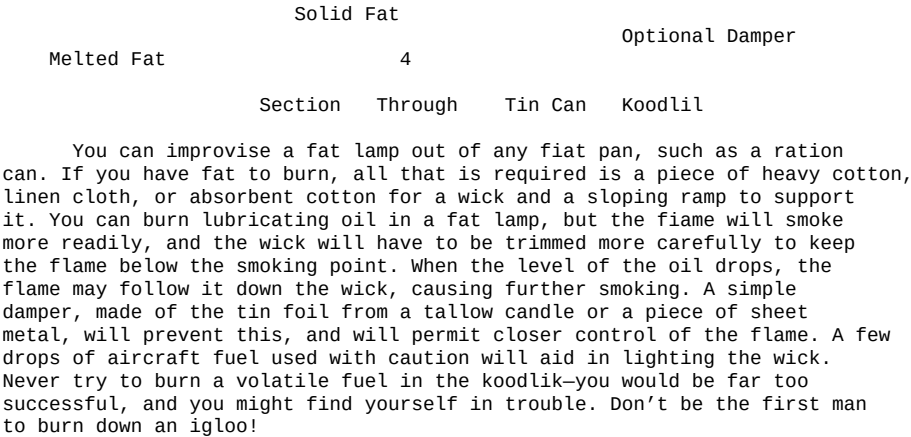
_ The left-hand men remain on the bench, assisting in cooking and maintaining their koodlik. If this lamp is burning animal fat it requires only moderate attention. Lubricating oil is not so easily used, as the flame smokes easily and the wick needs more frequent attention. A little animal fat dissolved in the lubricating oil makes a big improvement in the flame.

The Koodlik

The Eskimo fat burning lamp, or koodlik, has provided heat for comfort and cooking for thousands of years, giving a quiet and pleasant

SIGNALS

light and warmth to the native home. Properly tended it does not smoke or smell, and it can be controlled to give more or less heat on demand. It was carved laboriously from soap-stone in the form of a shallow pan of half-moon shape. The straight edge of the lamp was bevelled to support the wick, made of arctic cotton or moss. Seal oil or caribou fat was used as fuel. To avoid its melting into the snow shelf and to keep it warm enough to render fat, it was supported on short sticks driven into the shelf.



CHAPTER 6

SIGNALS

RADIO SIGNALS

Aircraft

Try the radio as soon as possible. Conserve the battery and be sure all switches are off after use. Power may be had from:

- (a) engines;
- (b) batteries – in winter, wrap the batteries in a parachute and keep close to a fire to keep from freezing; and
- (c) the APU – remove from the aircraft because of the danger of a spark igniting possible gas fumes.

Gibson Girl (AN/CRT3)

It is carried in the Aurora aircraft. Follow the instructions on the set. Power is by hand cranking, the unit held between the knees. Frequencies are 500 kcs and 8364 kcs.

On manual, the set transmits only on 500 kcs. On automatic, the set transmits six groups of SOS followed by a 20 second tone, and changes automatically from 500 kcs to 8364 kcs every 40 to 50 seconds. On "light" signal Must be keyed manually. There is no radio transmission when the switch is at the signal light. The maximum range on 500 kcs is from 320 to 480 km, on 8364 kcs from 1120 to 2400 km, The operating times are from 15 to 18 minutes and 45 to 48 minutes after each hour.

ELT (Emergency Locator Transmitter)

1. An emergency locator transmitter is a compact battery powered transmitter which sends out a distinctive signal on the distress frequencies of 121.5 MHz and/or 243 MHz.

2. There are four types of ELTs:

TYPE A – which automatically ejects from the aircraft and is set in "operation" by inertia sensors when the aircraft is subjected to a crash deceleration force.

TYPE F – a fixed type ELT which is set in operation by an inertia switch when the aircraft is subjected to crash deceleration forces. The transmitter can be manually activated and may be remotely controlled from the cockpit. It may also have an additional antenna for portable use. Type F is mandatory for all aircraft operating in Canada.

TYPE P – is similar to Type F except that it is not fixed and does not activate automatically.

TYPE W – is automatically activated by immersion in water. It has no fixed mounting but is buoyant and operates while floating on the surface of the water,

3. All four types must be equipped with batteries with a minimum operating life of at least 160 hours. To obtain the best range, the transmitter should be

placed on level ground and as high as possible so that there are no obstructions between it and the horizon. The antenna should be vertical to ensure optimum radiation of the signal. Placing the transmitter on a piece of metal or even the

wing of the aircraft if it is level will provide reflectivity and extend the end cap under the waterproof tape and strike the transmission range. Holding the transmitter close to the body in cold weather is

fusee. The spike at the other end is to stick into a not recommended since this procedure will not significantly increase battery life and the body will absorb most of the signal energy. burns 5 to 10 minutes bright red, but will not

4, In determining when to turn on an ELT, the survivor should keep in mind that an ELT signal could be picked up at anytime of day or night by the experi-

SMOKE AND LIGHT

mental Search and Rescue Satellite (SARSAT) now in orbit, also that an ELT signal can be picked up by high altitude aircraft from distances of over 160 km and that many commercial carriers and military aircraft listen out on distress frequencies, easy to use and very useful. Smoke is quencies, Also bear in mind that weather will not delay an ELT search so it is not calm days but works anytime. Fire and smoke a factor in deciding when to switch on the ELT. service.

5. It is recommended, therefore, that in almost all crashes or forced landings Pyrotechnic, Hand) that the ELT be turned on immediately after the aircraft comes to rest. If the

red star type flare ascends to an altitude of approximately 45 to 60 metres and burns approximately 7 to 15 seconds. it must be left on since turning it off can result in a search aircraft being thwarted. up and away from self and others when in its attempt to home on the transmission. firmly into launcher, nozzle end first, Ensure

6. Under no circumstances should the switching on of the ELT be delayed beyond 24 hours past the expiry of the flight notification or flight itinerary. projector at arm's length pointing it upwards, ease However, if an aircraft is seen or heard the ELT should immediately be switched channel and release smartly. on and left on,

PYROTECHNICS

Very Pistol – 3.8 cm

This is standard equipment in most military aircraft.

AP Day/Night Signal

This flare has two signals – orange smoke on one end for day and a bright red flare on the other end for night. Coloured smoke should not be used to colour snow unless more than enough smoke is available. Use it only when an aircraft is sighted. The smoke end may be used at night also; just light the smoke in the fire and it will produce a brilliant light. To operate, remove the plastic cap and fracture the seal by twisting the D-ring. Once the seal is fractured a sharp pull on the D-ring will ignite the signal.

RADA Parachute Flares

When fired, a rocket ascends to an altitude of from 370 to 460 metres. When at maximum altitude, a flare ignites and burns for a period of from 20 to 25 seconds. The flare is suspended by parachute. Operating instructions are clearly printed on the inner container, Follow the directions carefully. Hold the rocket firmly with one hand, pointing the rocket end skyward,

Fusees, Railroad Type
To light, remove the

flint against head of the

tree or the ground. It

burn if too wet.

Fire and Smoke

These are natural

excellent on clear fairly

will provide 24 hour

Miniflare (Projector

When fired, a single

proximately 45 to 60

Always point launcher

loading. Insert signal

signal is bottomed.

To fire, hold the

trigger knob to rear of

Method

(a) In accordance with the diagram, make three signal fires at least of the tree prepare a bonfire—using feather

30 metres apart if possible, and grouped in a triangle. Three splintered wood, bark, and any combustible material

oil from the aircraft if available).

fires, or signals in groups of three, are international distress calls.

(b) Along a creek bank or ravine, three fires in a line work. burn and ignite the "bird's nest" which helps

(c) Build signal fires in an open area, a field, marshy ground, or on tree making it a gigantic torch—visible to

rafts out on a lake or pond, if possible, and close to your shelter anyone for many km around.

so a quick dash can be made to light them when an aircraft is heard. signal in advance and use precautions against getting

or spreading fires.

(d) Have fires protected from rain and all ready to light with dry

oil are available, use only when the aircraft

feather sticks and splintered kindling in the centre. Place Jarger

the ground or saturate pieces of fabric and light

sticks around the kindling in tepee fashion and thatch with green

precautions.

boughs or moss. Keep additional fuel, green boughs, moss, and

very useful, important, and easy to carry piece

grass handy as emergency insurance.

has probably been responsible for effecting more

(e) If you are near the crashed aircraft, rubber and oil from it make

It is in nearly all types of survival kits. Follow

good black smoke.

of the mirror. It can be seen many km

(f) Smoke signal fires can be lit once a day to attract any possible day. If you have no mirror, improvise with a

local people in the area—forest ranger, campers, and rescuers,

side of a ration can make a good usable

who may see your smoke during the day or fires at night and

cross about two cm long in the centre of the

investigate.

knife while the metal is lying on a flat sur-

(g) Continuous burning is unnecessary and wasteful on wood and or candle light directed at the heliograph

energy.

(bh) If a low inversion above keeps the smoke in layers close to the carried in some aircraft, it can be operated from the

ground, it is generally possible to get the smoke above it by

building a larger fire prior to adding the smoke-making material.

Lights. Use these if they are still serviceable.

This applies to thick wooded areas, Find a good clearing away

aircraft, itself, is an excellent signal, Cut down

from trees if possible.

may conceal it from the air. Keep it clear of

(i) If a thin cloud layer exists, night fires will be diffused and not from the aircraft laid on the ground will

so readily spotted from the air. It is generally a good idea to

keep your signal fires going under such conditions anyway.

(k) Always be careful with fire and ensure that precautions are taken

AND COLOUR

SIGNALS

against your fires spreading.

Light Signals

quite effective when built in a clearing and of

Any form of light can be used. Camp fires, flashlights, candles, etc., contrast. In Canada a cross with arms running NE Sw

may be spotted from the air at night. A good trick is to employ light inside sun shadow.

a tepee which lights up like a giant Japanese lantern.

Torch Tree

Winter—snow block wall—line the blocks along the trench

(a) Select an evergreen tree with thick foliage, that is isolated from snow blocks were cut.

other trees to minimize the risk of forest fire.

Summer—sod, stones, sand, or driftwood walls.

(b) In winter, shake the tree or hammer the base and remove as

Winter—tramped in the snow; lay green bough signals in much snow and ice as possible.

(d) Around the base

sticks, dry

(like gas and

(e) This bonfire will

fire up the whole

aircraft and

(f) Prepare this

the kindling wet

Flash Fire. If gas and

is sighted. Pour the fuel on

them when needed. Take fire

Heliograph Mirror. A

of equipment, this device

rescues than other methods.

the instructions on the back

away on a clear sunny

piece of polished metal. The

mirror. Carefully cut a

sheet with the point of your

face. At night a flash-light

mirror may be effective.

Aldis Lamp. Still

aircraft power supply.

Aircraft Landing

Aircraft Parts. The

nearby trees and shrubs that

snow. Shiny pieces of metal

reflect the sun.

SHADOW

Shadow

Shadow signals are

sufficient size and

and NW SE will catch maximum

Construction

(a) Arctic

from which the

(b) Arctic

(c) Bush

better still stick them in snow and build a wall of
(c) Build a "bird's nest" in the lower branches of the tree using
boughs around them.
branches of other trees with dry kindling and bark mixed with
them.
Summer-rock pile signals-dead wood-bush or logs for
fresh peeled logs and bark or sod blocks.

the snow; or

brush and

(d) Bush

letters. Use

Colour

In kits are red or orange signalling cloth strips 20 cm by 3 metres

GROUND TO AIR SIGNALS -

which can be pinned to the open ground as an X or other emergency ground-air signal as shown on the plastic emergency parachute cards which most aircrew carry in their wallets. These cloth strips may also be waved strips of fabric, parachutes, peeled logs, stones, sod, branches in from the highest point around if an aircraft is heard. Snow can be coloured to provide maximum contrast. All figures should be at least 12 metres long. water sea marker dye is effective.

Use snow. Try

SOUND SIGNALS

INTERNATIONAL SYMBOLS CODE

These can be used to some effect among aircraft crew separated from each other during bail-out and help to guide a ground party to you or to

1

Require Assistance Vv

guide a hunter back to camp. Avoid firing three shots unless in trouble or effecting a rescue. Don't waste ammunition. Keep shots regularly spaced and in groups after ten-minute or longer periods. Shouting and whistling may

2

Require Medical Assistance . 1. x.

help guide rescuers to you.

INFORMATIVE SIGNALS

or Negative N

Leave a written note for a search party if you leave the crash scene or last camp site and are on the move. Write pertinent facts: date, direction travelling, number and condition of the party, etc. Leave the message in or Affirmative Y

4

a bottle or can, if possible, and suspended from a tree or tripod or under a rock cairn. A sign visible from the air should be laid out also to give the direction of travel. For yourself and searchers, blaze a trail.

5

Proceeding in this Direction -

Note. Check travel notes for blazing trails and leaving messages.

3

No

Ye

Am

CONCLUSION

Establishing contact with or attracting the attention of searchers and Tescuers should be your main objective as soon as your vital survival needs have been taken care of.

(a) Have your signals all ready and show them effectively.

a

(>) Prepare as many types of signals as you can at the best possible sites.

(c) Protect signals and equipment from moisture and cold.

Ae

(d) Remember that any unusual sign or colour contrast is visible from the air, even a single trail in the snow.

(e) Use pyrotechnics and guns sparingly and with caution—the supply in kits is limited.

(f) Smoke and a mirror are your best signals when no radio is available.

(g) Care for your signalling equipment—learn to use it and be found.

The following additional symbols are used by Canada only.

SYMBOLS		CODE	FIRE METHODS
ra			
6.	All is Well	LL	
7.	Require Food and Water	F	
8.	Require Fuel and Oil ; ;	L	
9.	Require Repairs	Ww	
NOTE:	Leave a space of three metres between symbols if possible.		

INTRODUCTION

Fire is one of the survivor's basic needs. With it he can keep warm, cook, dry clothing, and signal for help. No one should fly without having a means of lighting a fire in an emergency,

REQUIREMENTS FOR FIRE MAKING

- Spark
- The usual sequence for lighting a fire is from spark to tinder to fuel. The spark may be provided in one of the following ways.
- (a) Safety matches—these should be carried by all aircrew. Remember to stow a striking strip with the matches if they are packed in a waterproof container. "Strike Anywhere" matches are prohibited in aircraft.
 - (b) Cigarette lighter—a good source of spark as long as the flint and: fuel fast.
 - (c) Flint and steel—an easy and reliable method of fire lighting, If the sparking metal flint on the bottom of the fire tablet case is available, use a knife blade or similar piece of steel to scrape a spark from the flint into a tinder nest of cotton batting, scraped cotton cloth, or scraped paper fluff. A drop or two of gasoline will provide instant flame. If gasoline is not available, be sure that the tinder nest is large enough to ensure that the resultant fire will ignite to the kindling.
 - (d) Battery—an electric arc can be produced from the aircraft battery to ignite a fuel dampened rag. Don't do this in or near the aircraft!
 - (e) Pyrotechnics—an excellent source of fire. Caution must be exercised in their use, and you must beware of railroad fuzee or the night end of the AP produce an instant fire.

| : . ,
: ,
\ } '
needless waste. A
Signal will

Day/Night

(f) Magnifying glass—focus the sun's rays on a good tinder. The lens from a camera, binoculars, or reflector gun sight, or any convex lens should aid in fire starting. tinder to the fuel stage in fire lighting, it must

fuel materials require greater heat to ignite;
Metallic Flint on Fire Tablet Case
that some form of kindling be used to nurture
enough to ignite Jarger fuel. A few suggested forms
:
*
evergreen twigs;
shavings, wood chips, or fine splinters of resinous
(dry sticks shaved on the sides in a fan shape); or
oil impregnated wood.
should be gathered prior to attempting to light
maintain the fire. Different types of fuel are
requirements. Use what is available, bearing
woods burn better when dry and that pitchy woods or wet
wood is split the less smoky the fire will be.
Tinder hotter the fire and usually the slower burning.
The first starter should be fine, dry, highly flammable material such
as: but requires a hot fire to start. Split green
wood.
(a) cotton fuzz; * ° 4
(b) paper fuzz;
(c) absorbent cotton;
Ensure that the fire is well ventilated.
(d) arctic cotton grass tops;
(e) gasoline impregnated rags; or
{f} dry dead grass or witch's hair.
on mineral soil or solid rock. Forest fire
fires on muskeg, dry grass, leaves, evergreen
Nore.
water supply or sand is useful for
Tinder absorbs moisture readily from the atmosphere and may be
least effective when you most urgently require it.
Keep Your Tinder Dry!
scrape down to bare earth. In winter dig down
snow, or dig out an area around your shelter
exceptionally deep a small fire may be main-
layer of green logs.
is ideal if built on a gravel bar, presenting
depression because long logs may be
a fire directly under a tree because of the danger of
humus and leaves.
no value unless it is burning. Large logs
make an excellent burning reflector.
fire lighting are:
selection of tinder and fuel;
shield the match from the wind;
fire from downwind (or the leeward) side; and

Fuel

In going from the
be borne in mind that large
therefore, it is essential
the fire until it is hot
of kindling are:

- (a) dry, dead,
- (bd) birch bark,
- wood;
- (c) feather stick
- (d) gasoline or

A good supply of fuel
the tinder in order to
desirable for a variety of
in mind that all
woods smoke. The finer the
The denser the dry wood, the
Green wood will burn,
wood fine and start with dry

Ventilation
A fire requires oxygen.

Fire
Layout
The ideal camp fire site is

hazard is always present with
needles, or dead roots. A handy
extinguishing flames.
If the ground is dry,

to solid ground, trample the
and fire area. If the snow is
tained by lighting it on top of

A cooking fire on the trail
no fire hazard.

Avoid building the fire in
bridged up out of the hot coals,
Do not build
snow slides or igniting the dry

A reflector is of little c
rolled on the back of the fire

Four common mistakes in

- (a) poor
- (b) failure to
- (c) lighting th
- (d) smothering

the newly lit fire with too much fuel, too soon.

soaking wet and require an immediate fire there is a
the clothing will run down and put out the fire,
to keep it dry.
can be used to light damp kindling.
location sheltered from strong winds.
rocks around a fire. Some rocks will explode when
flying embers. You could lose all your equipment.
sleeping bag with para fabric or wool at night to prevent
bag.
night to permit easy lighting from the embers in
instructions carefully when using pressure stoves. Re-fuel
flames. Guard against carbon monoxide build-up.

4

ARCTIC

lack of natural fuels in the Arctic other than the odd
wood or stunted willow, and these are buried in winter.
carried or improvised.
cold weather drain oil from the aircraft before it
if necessary as it may be used in the solid state.
lamp may be manufactured as in the diagram. A
from cotton material, surgical gauze, or parts of
primed with a few drops of de-icer fluid or gasoline.
Kerosene. The remaining fuel in the tanks will burn in
or dirt filled can. Gasoline may be mixed with
Camp Fire Showing the Reflector log principle
Burn in a simple wick type lamp. A wick is placed
lid.
candles are in the emergency kit.
Seal oil or fat of other animals can be used as
fat lamp, using gasoline or de-icer fluid to start

Suggestions

(See Arctic Shelters.)

small tray of repellent with a sliver of wood or
Carry ample matches always.

will provide some heat. Do not waste repellent in
Keep matches dry. If damp, rub them through the hair or back and
forth between: the palms of the hands with the match head protruding
Dressings. These are in the first aid kit. Open at
slightly.

small piece as a wick, prime with a few drops of

Don't waste matches—use sticks from the fire for lighting smokes.
and light. After it has burned dry, the wrapper

Collect adequate tinder and fuel for the next day and keep them under
pieces of meat or fish. Fold the container air-tight. A
dry cover.

manufactured by bending the wire splint in the first aid kit

If the forest is wet use dry standing wood. Split the wood. Make

If you become
danger that water from
so hold the tinder high
The pressure stove
Pick a fire
Beware of using
heated.
Guard against
Cover your
sparks from burning the
Bank the fire at
the morning.
Follow
outside, away from open

There is a
piece of drift-
Fuel and stoves must be
Sources of Fuel
Oil (Engine). In
congeals—on the ground
A simple stove or fat
wick may be improvised
clothing. Oil should be
Gasoline or
primus stoves or a sand
oil to last longer.
De-icer Fluid.
through a tight fitting
Candles. Tallow
Seal Oil—Blubber.
a fuel. Make a simple

aoe.
rendering fat.
Insect Repellent.

match stick as 2 wick
summer.
Petrolatum Gauze
the centre, pull up a
insect repellent,
- used to cook small
grill may be
to any desired shape.

feather sticks for starting the fire.

Fuel
(oil, blubber, etc.

ie ig KOODLIK

AW

Pressure Stoves

The Primus (model 96) is the standard pressure stove in the military survival equipment at the present time. Its component parts are: the koodlik to support the wick.

Bend tin or aluminium and place in

Filler cap

Spirit cup

xZCAMOODD

Air valve

Flatten the top of the wire

splint stove to make a flat cooking surface. Wrap meat

Diagram of a Pump (with self locking valve)

wire splint

Head opening

stove.

Head

Flame cup

Vaporizing plate

Plug holder

or fish in foil from a candle.

Pe

In assembling, be sure that the head

(K)

Vaseline impregnated is screwed tightly into the Rauze.

This stove is built to burn coal oil (paraffin or kerosene) but it func-

with many types of liquid fuel. Never fill the stove over three-quarters

capacity and do so outside the shelter as a precaution against fire.

is set on a piece of metal or waterproof cloth one avoids wasting any

overflow. The head (K) has to be pre-heated in order that the fuel will vaporize to burn properly. This is done by filling the spirit cup with a readily combustible liquid (alcohol, naphtha, gasoline), which is ignited

allowed to burn almost entirely. At this time, the air valve (C) is

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left open. When the starter fluid is almost expended, close the air valve pump a few strokes (D) and the stove will light. If the starter fluid has burned out before the stove ignites, apply a lighted match at the top of the burner or head (K). If pumping is commenced too soon (before the head is warmed), the stove will flame up. Increase the pumping: decrease the flame by opening the air valve (C). To extinguish

flame by further

HUNTING

AND

leave the air valve open.

Warning. When fuel, such as naphtha or high octane, is being used, place a shield of cardboard, tin, etc., between the head and the air valve before opening the latter. If this is not done the flame will jump from the head to the escaping fumes at the air valve and this can result in extensive fire and possibly explosion of the stove.

HSING

If the flame does not increase after pumping or if the flame is streaky or not atmospheric blue, clean the opening (E) with the needles provided. Keep a lighted match near the head (K) when doing so in order that stove will re-light once the needle is removed. Handle the needle gently, especially in cold weather—it is easily broken. This cleaning process should be done as soon as it is noticed that the flame is not normal—if not, carbon monoxide can build up in an enclosed shelter.

When using the stove for frying, heating a shelter, or drying clothing it is advisable to place a piece of tin or other light metal on the uprights over the flame. This diffuses the heat.

The stove operates and lights best out of a wind or draft.

Carbon monoxide will be formed when insufficient oxygen in the air, Maintain ventilation.

remains

CHAPTER 8
HUNTING AND FISHING

. INTRODUCTION

The survivor must conserve and supplement his emergency rations in every way possible, so that he will not starve if rescue is delayed. In Canada this will be best achieved by hunting and fishing. Fish and game are usually available, and if some basic techniques are mastered the survivor can defer use of his survival rations for an indefinite period.

To the majority, hunting implies the use of the gun, but there are other productive methods of taking game. The survivor should plan to use every means at his disposal, favouring those which will require the least effort to achieve satisfactory results.

Snares, traps, gill-nets, and set lines will work for you day and night. Get them into operation as soon as possible. Utilize any available material and improvise. Scout the area of your landing early to learn its game potential. Tackle the job systematically.

Before departing establish a base line or check points by which you can always orient yourself in relation to your camp. This could be a river, a lake shore, a hill, or even a blazed trail north and south of your camp.

- The following general rules may prove useful to the hunter, +
- (a) Walk as quietly as possible.
 - (b) Move slowly, stop frequently, and listen.
 - (c) Look around.
 - (d) Hunt upwind or crosswind whenever possible.
 - (e) Blend with terrain features as much as possible, i.e., do not stand against the skyline or break from cover without thorough observation.

y

- (fE) Be prepared—game frequently startles the hunter or catches him off guard.

Watch for the following.

- (a) The animal itself—don't get excited when you see it; very often it isn't sure what you are and will remain still, Make all movements slowly and make the first shot count.

Dead sticks may be inserted

- * (b) Trails—usually beaten down, through heavy usage. If the rabbit

recently

the aa

used, trails are excellent for setting snares.

- (c) Tracks—may provide a wealth of information, such

as: the

type, size, age, and sex of the beast: the direction taken; the age of tracks, whether the animal was frightened; and so on.

- (d) Droppings—the best indication of what animal sometimes reveal favourite roosting spots of birds.

has passed; will

- (e) Feeding grounds, water-holes, and salt-licks—good locations for hunting in early morning or evening. Trails leading in to such places may be suitable sites for locating snares or traps.

- (f) Dens, holes, and food stores—good spots for setting snares.

'Commen Rabbit

Snare

(Using Wire}

a

SMALL GAME

General

'This end should de about

2kg heavier then the other end. Make certain that the

The mainstay of the survivor, particularly if he has no rifle, will balance pole will lift the

probably be small animals and birds. These are well distributed through the Canadian hinterland and may be taken without firearms.

Rabbit

There are several species common in Canada. In woodlands they frequent heavy thickets. They are taken in snares set on their runways, preferably where the width of the runway is restricted by natural or man made obstacles. The balance pole snare and the common rabbit snare are shown in the diagrams.

Balance Pole Snare
(Using cord)

Rabbit mares should non
mally de 12 cm in diameter
and 8 cm from the ground.

Squirrels

These are common throughout Canada's forests. They store their food in tree cavities, nests, or holes in the ground but their provender is seldom suitable for human consumption, consisting mainly of pine and overlooked by the survivor. spruce cones and the occasional mushroom. The leaning pole snare is a simple and effective method of taking squirrels. It should be employed near their food caches or nests. Three or more snares to a pole are forested areas. Watch for tree with the desirable, since squirrels are fond of the company of other squirrels. ground. Its main food is generally

with a club or spear. Be careful

skin, open the hide along the

of the quills. Work from the

contact with the quills.

Squirrel Snare

Fur Bearers

delicacies but if you can shoot, trap, or

a

and marsh areas; the beaver

poplar or willow country (watch

and the best time is:

musk-rats maintain paths and

evidence of these creatures coming

early spring should be checked;

water.

|

plentiful, except by hunters em-

palatable of the carnivores.

pork.

except when emboldened by hunger. How-

their bark or howl. Fox can be

fox can be captured in a baited

point as the fox usually travels

Mice and Lemming

These are edible and should not be

Porcupine

The porcupine is found in most

bark stripped off fairly high above the

bark from trees. It can easily be killed

in handling—pick it up by a front foot. To

belly and peel the hair back over the top

inside as much as possible to prevent

Mink, Marten, and other Carnivorous

These are not rated as table

snare them put them in the pot.

Musk-rat and Beaver

The musk-rat is found in pond, slough

along streams and lakes passing through

for dams). They may be shot while swimming

early morning or late evening. Beavers and

runways where snares can be tried. Any

out from under the ice in the late fall or

you may cut one off from the safety of the

Lynx

Seldom seen, even when they are

employing tracking dogs, the lynx is the most

The meat is not unlike tender, lean young

Fox, Wolf, and Coyote

These are seldom seen

ever their presence is often revealed by

captured by a bait snare. In the Arctic,

beehive trap; locate the trap on a high

over rather than around hills.

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Beehive Trap

Constructed from piled up

rocks and stones.

Balance pole tied to
 Sdjacent tree Ensure that
 the weighted balance pole
 is balanced and positioned
 '30 that it will lift the
 vielim clear of the ground.
 A Cord with moose.
 B Cord to teigect
 mechanism.

Front View

Dotted line represents the
 fence which ensures that the
 animal must insect its.
 head through the noose to
 'obtain the bait,

along most of the arctic coastline, is an
 capture and from it obtain food and fuel
 underwater and comes up every minute or
 will swim several hundred metres before
 June it will float; hence, if within
 retrieved. During the remaining
 shot, due to its lean body condition and
 inshore coastal waters at that time. In

Seal

The Common Jar Seal, found
 animal which the survivor may
 (blubber). It generally swims
 two to breathe. If frightened it
 surfacing. From the end of August t
 range it can be shot in the head an
 three-month period it will sink, if
 the decreased salinity of arctic

open water, it is quite curious and will come in close to investigate a scratching or tapping sound.

During the winter the Common Jar Seal lives under the ice and maintains open breathing holes, which are usually covered with snow. These holes are not easily found by the novice, as they are usually located in pressure cracks off points of land. The seal is captured here by waiting without motion until it surfaces and then shooting it in the head or by setting a seal hook at the bottom of the hole (underside of the ice) on which it will catch itself upon diving.

In the spring the Common Jar Seal can be stalked as it suns itself on a perch which rests

the ice. It will raise its head every minute or so, and at this time the stalker should remain motionless. If white clothing is not available for camouflage, the stalker should crawl and slither within shooting range. Bevelled the end edge against of the aperture.

BIRDS

Upland Game Birds

'Weight—Hang this low enough so that birds will not land on it

Grouse and partridge are most often found roosting in thickets, instead of the perch. sunning on side slopes, or feeding on the ground, although they are occasionally found in trees, roosting or feeding on buds. Silence in hunting 'The eperue and perch may be them is not essential. Normally these birds will not fly very far when round a fence to construct, and

the aperture may be nearer the top in the tapering part of the post.

frightened and, therefore, it is better to flush them out and wait for them to land before shooting.

The spruce grouse in the woods and the ptarmigan in the Arctic 'Skat: Retains the perch in position until the landing bird depresses the perch, allowing the weight to pull the knot through the aperture thus trapping frequently remain motionless thereby seeking to avoid detection. This habit the bird by the feet in the noose. often plays them into the hands of even the most ill equipped hunter.

Ojibwa Bird Snare

They are not difficult to take with a catapult, many are killed by thrown sticks and rocks, and the spruce grouse is easily garrotted, at times, by a

BIG GAME

wire noose on the end of a pole.

General

Water Fowl

period, but it

A big game animal will provide food for a prolonged Birds such as ducks, geese, and coots are normally more difficult to will require a suitable gun and some skill, and it normally entails a

approach than upland game. It is therefore advisable to shoot them on relatively large expenditure of energy to secure one. If, when scouting

the water, taking into consideration the way the wind and current will the area, you encounter fresh signs of big game you may plan a hunt. take them. If geese are found in their nesting areas during the moulting Most big game is abroad at dawn and toward evening. The dawn period, it may be possible to run them to the ground. Do not overlook day

hunt is best for the survivor. If he loses his bearings, he will have all to

the eggs or the young. If any waterfowl taken are suspected of being fish to find his way back to camp. If he gets game, he will have daylight

eaters, it is advisable to skin rather than pluck them. This, of course, dress it and begin preservation of the meat. depends upon your tastes which will be influenced by your appetite. ee"—

:
The Deer Family

while

Other Birds

Deer and moose are found throughout Canada's forest zone

All Canadian birds are edible and most are palatable, but do not woodl and caribou are most common in Western Canada. ; cik and

ng

expend valuable ammunition for little gain. Gulls will often take a baited In summer, follow ridges overlooking open country, but avoid showing hook. An Ojibwa Bird Snare will be quite effective in capturing the Canada yourself against a skyline. Look for salt-licks and wallows. Flies and ticks ing

Jay (Whiskey Jack or Camp Robber) which will visit most woods camp
torment these beasts during hot weather, and at times their frantic splash
a distance. Watch game trails, because most
animals
sites, and a few of them can be made into a stew.
can be heard at
prefer to use them when travelling. In winter, the deer, elk, and
moose

tend to "yard up" in low lying sheltered places, such as cedar swamps,
The gut region is usually messy but is not to be overlooked if no
willow clumps, alder swales, or poplar thickets.
other shot is available. The wounded animal may travel for many
When stalking game, hunt upwind or crosswind. Avoid any noise, and
kilometres, leaving no blood trail, before it dies.
stand motionless frequently, scanning the area. Most game is seen while
When game is taken, bleed, clean, and cool it at once to avoid spoilage.
the hunter is standing still. Don't go crashing across country.
Still hunting requires less energy and is often productive. Wait quietly
Snaring
and remain motionless in spots overlooking meadows, game trails, salt-
Snares set on well worn trails may save many tedious hours of
licks, and wallows, preferably crosswind from the area under observation.
walking or waiting.
If an animal suddenly starts up ahead of you, do not give chase
A snare of cable or heavy wire 60 cm in diameter and suspended
immediately, It may not have known what you are and curiosity may bring
approximately 45 cm above the ground should produce good results.
it back to check. If you remain motionless and keep watching on all sides,
If the snare is well anchored, the animal will probably kill itself in a
you may get a good opportunity to shoot it. Deer, especially mule deer,
short time.
do this quite frequently.
Should you wound one of these animals, do not follow it immedi-
ately. If it knows it is being followed it may run many kilometres be-
fore it drops, but if you wait five or ten minutes or long enough to
smoke a cigarette, it may lie down after a short run and either bleed to
death or begin to stiffen up. If you wait too long the animal may regain
its wind and depart before you arrive.

weeny

Where to Shoot your Big Game

The head or neck area presents a small target unless the animal is
close. It may move its head and neck as you shoot.

The heart and lung area is the best shot for most hunters. It is quick
and certain, and the animal will not go far.

A Method of Snaring Antlered Game

Use a strong, dry pole about 3 metres long and 15 cm in diameter.
Apache Foot Snare

The Apache foot snare is another effective snare that may produce
results when used on a well travelled game trail.

A hole the same diameter as the width of the game trail and about
15 cm deep is dug in the game trail. A rectangular piece of heavy
paper or cardboard or thin aircraft skin with a 30 cm cross (X) cut
in it is laid over the hole. A snare made of shroud line is placed over the
cardboard and fastened to a heavy jog. The set should then be camou-
flaged with light sprinklings of leaves and earth.

Typical Target Area When Shooting Big Game

The cardboard or paper will ensure that the snare remains on the
Mountain Goat
animal's foot until it is drawn taut. The animal will be able to drag the
log until it is exhausted, then it can be caught and killed.
only at high altitudes. If an opportunity of hunting

approach them from above. Often these animals are
falling from a precipitous location. When skinning
having the hair touch the meat—it will taint it.

FISHING

most effective in still water, e.g., a lake (near the inlet
locations) or back water in a large stream (for
to block the stream). Nets can be constructed
of parachute shroud lines. The floats on top and the
are to keep the net vertical in the water. When
fish are inclined to stay deeper. The smaller the
fish you can catch, but a small, mesh will still
A mesh of 7 cm is a good standard.

Barren Land Caribou

Once the herd is located, the caribou
reareareeanmiennens
SOON

particularly when migrating. Approach them
SOD

a shot is fired they will often flank you quite
are. When grazing in small herds, usually
watch the horizon; when hunting them do
upright position, Caribou meat is delicious.

are usually easy to shoot,

low and upwind; even after

closely to discover what you
in the valley bottoms, they
not top hills quickly in an

Bears

Most bears are not dangerous unless injured or provoked. Bears do
become irritable with age, however, particularly grizzly bears, and the
survivor who hunts them should be well armed and cautious. Be on guard
if you spot cubs as the mother will seldom stray far from her youngsters
and she will fiercely resent any intrusion upon her domestic affairs.

Musk-ox

Gove Hite

'Qverhasd Kaot

Musk-ox may be encountered on most of the Arctic Islands and
portions of the Arctic main land. When danger approaches they face out
in a ring around the calves. A gun is required. Approaching extinction,
they should not be molested except when in dire need.

Mountain Sheep and

These are found
them is presented,
shot but lost through
mountain goat avoid

Netting

A gill-net is
and outlet are good
survival don't hesitate
using the inner cords
weights on the bottom
ice is on the lake, th
mesh, the smaller the
entangle a large fish.

Here are two methods of setting a net without the help of a raft or boat.

Summer-setting the net out from the shore with the aid of a long Refraction can be counteracted by keeping pole.

- while waiting. Along the arctic coast
- Anchor Line-Pulls the net into place.
- within spear range by bobbing a shiny

of wire attached to a long pole. The head and the fish is then jerked from the

of fishing which does not require the lines may be set, using as many baited Gill Net Set for Summer Use parts of fish (eyes, fins, head, or strips of

red meat are all good baits. One method Winter-setting the net by cutting holes in the ice on a lake. Ensure stomach of the first fish caught to the net is set several centimetres below the ice to prevent it from freezing.

*

wobblers) moving; it attracts the fish more another technique. Here a short stick, near the hook are employed-the last the attention of the fish.

Gill Net Set for Winter Use

With a pole slightly longer than the distance between the holes in the ice, attach a line to one end and, starting at hole A, float the pole to B, to C, and remove from the water at C. Attach the net to the end of the line and pull the net through A until it is set, as shown. Ensure that the line is tied to both ends of the net, to assist in checking and resetting.

Method of Setting Gitl Net Under Ice

Spearing

Clear water is a requisite. the tip of the spear in the water the arctic char can be attracted object up and down in the water,

Snaring

The snare consists of a loop Joop is passed over the fish's water.

Hook and Line-Set Lines This is an effortless method

presence of the fisherman. Night hooks as possible. Insects, worms, belly) or whole bloody fish, and of selecting a bait is to check the discover what it has eaten.

Trolling or Casting

Keep the bait (spinners or Teadily. Jigging through the ice in a fishing line, and a shiny object being kept on the move to attract

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FOOD AND
WATER

CHAPTER 9

FOOD AND WATER

INTRODUCTION

Food is not, as many people think, an immediate requisite of survival. Man can exist for some time on nothing but water and his own body fat.

The daily caloric requirement of a person varies considerably, depending on age, weight, sex, etc., but is always directly proportional to the amount of energy expended by the body. It is likely that the daily caloric requirement of an average person doing an average job would be somewhere in the neighbourhood of 3,000 calories.

Experiment indicates that a healthy person can subsist for a considerable period of time on 500 calories a day without harmful effects. The aircraft emergency ration pack, when used according to directions, will provide the survivor with approximately 500 calories a day.

Vitamin C cannot be stored by the body but fortunately it is usually obtainable under survival conditions. It usually takes several weeks of improper dieting to bring on scurvy symptoms.

Under conditions of strenuous exertion or cold, additional caloric intake is needed to maintain body temperature. The ration pack is inadequate for this purpose, and food should be supplemented in any way possible. If game is readily obtained it is possible to subsist indefinitely in good health on a diet of meat, fat, and water. When other foods are available, use them in preference to the emergency ration which should be regarded as a last resort.

WATER

Water is more necessary to human existence than food. About half a litre (two cups) is considered to be the normal minimum daily requirement. Obtaining good water should present no problem in any of the sparsely settled areas of Canada. There are no known sources of deadly poisonous water. Contamination is normally found only in the vicinity of human habitations, where the practice of survival techniques is seldom required. Do not drink sea-water. It will increase your thirst

and "tnay cause further loss of body fluids through diarrhoea and : body if the snow is excessively cold. If heat is not available melt small quantities in your hand by vomiting.

squeezing and breathing on it before consuming it.

Summer Water Sources

Spring water or fast running water is best but any running water or that from properly drained lakes in isolated areas will be safe.

and shake afterward to restore oxygen and ~

Standing water of sloughs and muskeg areas can be used with careful bailing. In muskeg areas where the growth is in mounds of varying heights directed.

you will often find small pools of good water around the bases of the per litre. mounds.

sediment in water, use filtering or settling

Sometimes when surface water cannot be procured, water can be obtained by digging down into moist soil (usually in the low ground of

unpalatable water may be improved by depressions, gullies, etc.). Muddy water may seep in but it will become

and allowing it to stand overnight. clearer if allowed to settle.

Sea-water can be used if a desalter kit is available.

Pools of good snow water can be found on the sea ice in late spring.

FOOD

Other sources are the sap layer of trees such as birch and maple in spring, dew on plants, and rain-water. recognized as a universal survival ration.

approximately 500 calories a day.

'Winter Water Sources

place a maximum amount of the best universal If open water can be found or water obtained through ice, it is to space provided in jet cockpits. The term be preferred to ice or snow as no heat is required.

ration" means that the most palatable ration which

It requires approximately 50 per cent less fuel to obtain a given water and which still produces adequate quantity of water from ice than it does from snow.

with few ill effects if a minimum of effort

A pointed instrument is best for breaking out ice—a number of light following information should clearly illustrate taps to start a crack, then one sharp tap to break off a chunk of the size been adopted.

required. On a large surface of a lake or stream, cut toward a crack which

yields approximately four calories is already there in order to avoid getting only splinters and spray.

of heat energy.

Icebergs frozen in arctic sea ice are the best source if obtainable.

water) —yields approximately eight calories The thin layer of frozen salt water spray outside should be chipped off.

of heat energy.

There is usually dangerous thin ice or open water around icebergs caused by tides and currents acting on the bulk of the berg which is below the

yields four calories of heat energy water.

Salt-free sea ice can be found where the ice has summered and

(plus water).

frozen in again—usually along the tops of ridges where the salt has leached be consumed in accordance with the instruction. It is bluish with a crystalline structure in comparison with salty ice During the first day when no food is taken, which is grey and opaque.

body fat stored from previous eating. It is

Snow which has lain on sea ice for a period of time usually contains physical work, such as building shelters, salt.

fires, etc., as soon as possible during the first

Boil three to five minutes

eliminate the flat taste.

Use Halazone tablets as

Use eight drops of iodine

If there is considerable

processes.

The flavour of safe but

adding charcoal from the fire

Emergency Ration Pack

The carbohydrate ration is

It provides the survivor with

It is designed to

survival ration in the limited

"best universal survival

can be eaten with little or no

energy to keep the body alive

is expended. A glance at the

why the carbohydrate ration has

One gram of protein (plus

water)

One gram of fat (plus

but

One gram of carbohydrate

(when assimilated by the

body)

The entire ration should

tions enclosed in each can.

energy will be derived from

advisable to do as much of the

collecting fuel, laying signal

Hard packed snow yields more water than fluffy snow.
available.

If fluffy snow must be used then pack it down into the container. In carbohydrate, but there is sufficient fat to trigger either case once the container is over heat work the snow with a knife the stomach contractions which produce the or other such instrument until there is more water on the bottom than containers can be used to melt snow or ice, make will be absorbed by the snow above it. This will prevent the bottom of provided with wire handles for such purposes. the container from becoming dry and burning out and also the resulting flying units have authority to place on their water from having a burnt taste. Do not eat snow. It tends to dehydrate the have carried for emergency use.

day when maximum energy is

The ration is mainly juices which will counteract sensation of hunger. The coffee, etc., and some are

Commanding officers of aircraft any foods they wish t

Natural Food

is found in the wooded areas of
Every effort should be made to discover and obtain natural foods indistinguishable when young but in:
within the crash area before the emergency rations are expended. This cup-like formation at the base and a search is, of course, going to be in direct ratio to available energy. part way up the stem. Avoid al! mush-
Naturally the two sources are plant and animal life.
structure and also young mushrooms in the
Plant Life
been confused with the common edible

It is difficult to provide comprehensive instruction on plant recognition in a short period of time. Here only the general rules and plants to be definitely avoided will be covered.

Most green plants are a potential source of Vitamin C-rose hips (the buds of the wild rose), tea made by pouring boiling water over labrador tea leaves, spruce tips, willow tips, dandelion leaves, etc.

Anything that is not bitter or anything eaten by birds and animals is probably, but not necessarily, safe to eat. If doubtful, take minute quantities at first and wait 24 hours for a reaction. Gradually increase the quantities, using the same test period, until certain.

Poisonous Plants. There are no poisonous plants north of the tree-line, However south of the tree-line there are three which can be mistaken for edible plants and can cause death.

- (a) Water Hemlock. 60 to 120 cm tall, it is the most poisonous plant in Canada. It is a member of the carrot family and has toothed three-part purple streaked leaves which emit a disagreeable odour when crushed and hollow tuberlike roots which emit a parsniplike odour. It could be easily confused with Cow Parsnip which is edible,

Water Hemlock
(Cicuta Mackenziana)

(b) Death Cup Mushroom. This northern Canada. It is maturity has a soft white broad collar-like ring rooms having this button stage. It has mushroom by some.

Death Cup Mushroom
(Amanita Phatlolds)
(usually 10 to 15 centi-

metres tal] when
mature)

Note the basal cup

Over mature
mushroom

Mature
mushroom

perennial 60 to 90 centimetres tall
in a short thick terminal cluster. Red
the flowers and resemble dolls' eyes in
Leaves and
Flowers
is substantial. Avoid berries grow-
type.

Air or Water
Pockets 122

123

(c) Baneberry. This is a bush with small white flowers or white berries replace appearance. The root-stal ing in clusters of this

These are dry scale-like plants, usually found on rocks
stumps in both the Arctic and woods. They can be bailed
then ground into a powder and stewed for use as a
thickener, gravy base, and mild seasoning. A common
lichen of the Boreal Region is the lemon lichen.

Although some mushrooms are edible, delicious, and
they are not particularly high in nourishment and since
extremely poisonous, they should be left strictly alone
positively identified as being edible.

leaves of labrador tea, a shrub-like growth found

practically all muskeg areas in Canada, can be steeped to

aromatic stimulating beverage. These leaves are boat

brown pubescent underside. This infusion is quite

vitamin C.

from many trees, such as the maple, bass-wood, birch,

is a source of water with some food value particularly

spring of the year. At other times the cambium or

growth between the wood and the bark of such

as poplar and jack pine can be scraped into a

this pulpy residue is tasty and nourishing and contains

(e) Lichens.
or old
and dried,
soup
edible

(f) Mushrooms.
filling,
some are
unless

(g) Leaves. The
on
produce an
shaped with
high in

(h) Trees. Sap
and poplar,
during the
succulent ne
common trees
container,
vitamin C.

Edible Plants

verses and rules which supposedly differentiate between
(a) Flowers of most plants in Canada are safe to eat either boiled
poisonous berries and mushrooms but none are com-
or raw.

rule is for the survival student to become familiar
(b) Greens such as dandelion leaves will provide roughage and
can be used as variations in diet and a supplement
vitamins. These can be eaten raw or stewed. Some delicious
The best place to do this is on field trips. Only
edible greens are: bracken fronds or fiddleheads, the lower
lifetime study can hope to exist satisfactorily for
tender inner twelve inches of cat-tail or bulrush stalks, cow
on plants alone.
parsnip, young green milkweed pods, young water-lily seed pods,
illustrations of edible plants see Appendix A to this chapter.
and pigweed.

(c) Roots such as cat-tail, wild carrot, liquorice, etc., will provide
starch and protein. Cat-tails are an excellent source of starch
locusts are considered to be a delicacy in many
since they are found in most parts of Canada and are obtainable
remove the wings and legs and toast the body on a
in both winter and summer. They are best boiled. Some other
at all objectionable and is said to be nut-like.
common delicious edible roots of Canada are: bracken roots, vetch
wood burrowing larvae of beetles, are usually found
roots, tiger-lily roots, lady's slipper roots, and the tubers of the
Toast or boil them.
arrowhead plant. Water-lily roots and jack-in-the-pulpit roots are
especially the large black type often found in
edible but must be boiled in two changes of water to remove
head, thorax, and legs and eat. The eggs are
acrid flavour.
procured but are dry and tasteless.

(a) Berries. The edible berries of this country are too numerous
to list here. Blue and black berries not in clusters are generally
safe. Red berries are more likely to be unsafe and white berries
terrestrial snails are an excellent source of nourish-
should be avoided unless positively identified to be safe.

There are many
poisonous and non-
pletely true. The safest
with a few plants which
to the emergency rations.
those who make this a
a prolonged period of time

For

Insects
Grasshoppers or
countries. It is best to
stick. The flavour is not
Grubs, the white
in rotting logs or stumps
Ants are delicious,
the rotten logs. Remove the
edible if enough can be

Snails
Both aquatic and
ment when obtainable.

Earthworms

BUTCHERING

The large types particularly are eaten in many countries.

Reptiles

Lizards, frogs, snakes, and turtles are all exceptionally tasty whether boiled or fried,

Game—Meat

All Canadian birds and animals are edible except the livers of the polar bear and bearded seal. These have an excessive vitamin A content which produces a toxic reaction.

Dressing Game

Some birds, such as ptarmigan and the various species of grouse, can be skinned using the fingers only. Break the skin on the breastbone and work around the body. Sea birds such as guils, fish-eating ducks, etc., should normally be skinned to diminish the fishy taste. The craw or first stomach should always be checked as the content is not digested and may provide nutritious foods, such as buds, berries, seeds, etc., which can be eaten.

The following diagram will provide a good general pattern for skinning most animals. When the animal is too big to elevate, skin off one side to the centre of the back, then spread that skin out on the ground to protect the meat while you roll the animal over and skin the other side. AIL

1, Rump 2. Round 3. Loin 4. Ribs
internal organs of game should be removed immediately to prevent bloating.

5. Chuck 6. Neck 7. Flank 8. Short Ribs
Big game is usually bled out by the bullet which kills it; however, it is 9. Shoulder 10. Shank.

advisable to cut the jugular vein right after killing to assure thorough bleeding, hence better tasting and better keeping meat. The removal of internal organs from small game and birds may be left until return to camp.

Big game is more easily utilized when it has been butchered into head manageable chunks as shown here.

cavity cut to A

Heart, liver, and kidneys of all game can be used (except as indicated the shortest distance from the cavity cut to C. above) and the eyes, brains, tongues, stomach lining, and even the stomach legs at C.

content of hooved animals can be consumed for roughage and vitamin content. Marrow of bones is also good. Be careful not to break the gall sack when removing the liver. Incidentally, there is no gall sack in deer or other antlered game. In hooved animals all parts can be eaten. The same rule applies generally to small game.

COOKING OF MEAT AND FISH

method of preparing meat for human consumption.

less fuel than other methods, and, if you drink the

you get the full food value. A small quantity will

and there is no waste. In order to assure all

have the chunks of equal size,

Sticking point
(for bleeding)

A Remove the

B Extend the

C Cut the hide

Cut off the

Boiling

This is the best

It is easy, it requires

resultant broth as well

feed a number of people

meat is properly done,

Broiling and Barbecueing or Roasting

be held to the side of hot coals not over them. Avoid smoke and flame. Use a receptacle to catch drippings.

22 eee a aa
Plank fish or meat on a split log with pegs. 4

A simple method of suspending the pot over the fire.

ide ee _

weight on the butt of the pole if necessary. The pole can be rotated

BOILING

to baste all sides of the meat.

Frying and Boiling

These can be done if the utensils are available and if there is sufficient the Lone Survivor

food. A certain amount of the food value is wasted, and above the tree- the lone survivor, who may find himself without cooking utensils, line, where you may have only a limited quantity of fuel, you will waste steaming, and barbecueing are good methods to use.

fuel.

bake, hollow out a shallow pit and line it with rocks. Place over

bed of coals and on it place the food wrapped in sticky clay

Barbecueing or Roasting

Meat, fish, and tubers can all be cooked by this method: the first

This is an easy method and often produces the tastiest results though in small pieces, e.g., a rabbit's leg. Over the food place more it causes the most waste. Clean the fish or small animal and spike it on and cover the whole with a few centimetres of earth. Leave the end of a green stick elevated beside a hot fire, which produces as or so until cooked. The pit does not have to be lined with little smoke and flame as possible. If necessary turn the meat occasionally foods cook faster and more thoroughly when they are used.

to assure it is thoroughly cooked. A great quantity of nutritious juices is steam, use the same rock lined pit but after burning the fire to lost by this method and there is considerable shrinkage. However this can

bed of coals, scrape the coals out and replace them with wet be partially counteracted by either placing the meat very close to the fire

grass. The food is wrapped as before (tin foil from rations is

dampened parachute material) and covered with a second at first to form a hard crust on the outside which will contain the juices, or having the meat suspended at one end of the fire with a shallow plate grass, then earth. Punch a hole down to the food level and pour underneath to catch the drippings.

quantity of water. Then block the hole and leave for an hour

let the steam from the water on the hot rocks cook the food.

Meat should

AM

Use a

Cooking by

For

baking,

To

them a good

or mud,

should be

hot coals

for an hour

Ticks, but

To

achieve the

or green

good—even

layer of

in a small

or so to

It is inadvisable for the survivor to eat meats and fish meat away; cut or scrape off the mould there is a complete lack of heat. Fish and many animals may tundra areas dig to the perma-frost or hosts to tapeworms, flukes, etc., which would be harmful cooling is necessary. but thorough cooking, e.g., boiling for 20 minutes, will kill disease germ and all parasites.

sun, wind, etc, can be used to preserve. Larger

trees above the fly line (the line

not found due to air turbulence) and the

crust which seals and protects the inner

that separate pieces of meat are not

will commence any place where the outer

Fish and meat can be filleted and spread'

the sun and wind. This is called

bright sunny weather. Meat can be best

about twelve cm by five cm by half acm

Smoking

meat and fish deteriorate rapidly it
A Simple Method of Baking Food.
meat by smoke-curing. Smoking not only

flies and like "jerking" it dehydrates and
PRESERVING OF MEATS AND FISH

half a centimetre thick, on a green-
To have a supply of food in excess of daily consumption

and one half metres above a slow-
objective of the survivor and if achieved, he must take steps to maintain
the supply by avoiding spoilage and theft by small animals.

birch, and dwarf birch are suitable smok-
Freezing

such as spruce or pine, should not

In the winter this is no great problem; the food is frozen solid and
blackens the meat and give it a disagreeable
stored in a place not accessible to animals, One thing to remember is
that meat will spoil if thawed and re-frozen, particularly if this happens a
smoke house to obtain a concentration
number of times. Before freezing cut it into pieces of a size that can be
used at one time and keep it frozen until ready to use.
smoke. Do not let the heat cook the

juices. A sod-bank tunnel is a means of
Cooling

provide a cooler supply of smoke.

In the summer, small quantities which are to be kept for only a day

the meat is brittle. It can then be
or so, should remain as cool as possible. If a metal container with a lid

desired, but avoid eating raw, smoked
is available, set it in water or bury it in damp earth, preferably in a

may contain parasites.
shaded location. One method of cooling which has been successful is
digging a hole in a river bank, lake shore, or hill side and blocking the
entrance with a large piece of sod. Added cooling can be achieved by
replacing the sod blocked door with a curtain of the heaviest possible
material. This is saturated with water each morning and the evaporation
process during the day will cool the chamber behind. The metal, or even
wooden, container is to keep rodents from the food-the container
should be ventilated. Remember that if it is excessively damp, meat will

raw unless
be diseased
to humans,
almost any

soon mould. Do not throw mouldy
and cook as usual. In arctic and
within 30 to 60 cm; no additional

Curing

The out-of-doors itself,
pieces of meat can be suspended i
above which insects are usually
outer layer will dry into a hard
meat. In using this method be sur
touching each other as rotting
surface is not completely exposed
out on willow racks to be cured b
"jerking" and is effective only i
jerked by cutting it into strips
(or thinner).

Preserving Meat or Fish

In warm or damp weather when
may be advisable to preserve the
preserves the meat but also deter
reduces weight.

(a) Lay strips of lean meat

wood grate one to one

burning fire.

(b) Willow, alder, poplar,

ing woods, Resinous woods

be used as they will

flavour.

(c) A tepee can be used as a

of smoke.

(d) Avoid excessive heat and

meat or draw out the

isolating the fire to

(e) Continue smoking until

eaten raw or cooked as

fresh water fish as the

tail	Cat-
Latifolia)	(Typha
or wet ground. The	Grows in marshy
edible when young	flowering head j
asparagus.	and green. Boil like
portion of the	The white succulent inner
is widely used	lower 30 to 37 cm of stem
	as food. Boil or eat raw.
or roasted.	Roots may be baked, boiled
the starchy	If woody, chew and swallow
residue.	material. Spit out the
'Nutgrass	
A semi-isolated tree is less attractive to small climbers. Suspend the meat above	(Apios Apios or
Apios Tuberosa)	
the fly line and out of the reach of leaping animals, A piece of parachute material	
will protect the meat from the elements but leave it loose enough to allow free	
stem and grows in	Has a three-angled
air circulation.	low ground near
water.	
Caches	
eat raw or boiled.	Peel the root and
Surplus food and supplies should be cached. There are numerous	(Better boiled.)
methods of doing this. The best is to suspend the meat by rope, away	
(Apois Americana)	The Ground Nut
from trees or uprights. The main threat to your cache is animals, from	
similar manner.	may be used in a
the bear to the mouse. The cache should be high enough to prevent attack	
from the ground, three to five metres, and far enough out from the	
uprights that climbing and springing animals will have difficulty reaching	
it. Squirrels and Canada Jays are the nuisances. Food can be covered to	
ward off meat-hungry birds, but from the preservation point of view it is	
advisable to have the cover fit loosely so that air can circulate around it,	
To do this have the cover also suspended by the rope and hanging around	
the meat. Have drying racks fairly high and if possible wrap the uprights	
with metal to stop climbers. Avoid overhanging branches.	

Lichens

There are many varieties of lichens which can be eaten. Some can be nibbled raw but they are generally acidic and should be soaked in water for several hours then removed, dried, and crumbled, before boiling to a gelatinous consistency. May be used effectively in thickening soups, stews, etc., or as a broth.

Reindeer-Moss
(Cladonia-Rangiferina)

A greyish, coral-like plant. Can be powdered and mixed with flour, or boiled to make soup.

The
substitute
substitute for

5-15 cm high

Dandelion
(Taraxacum Officinale)
The leaves are an excellent green.
dried, ground-up roots make a
for chicory which is a
coffee.

Bracken
(Pteridophytes)

Cow Parsnip
(Heracleum Lanatum)
Young leaves and flower stalks make a
sweet green. Eat raw or boil.

Rootstalks are woody. Roast are nutritious but quite or boil.

Fiddleheads or young curled up fern
fronds make a delectable green when
boiled.

The epithelium or tender growing por-
tion at the base of the stem is good

Liquorice Root
either raw or boiled.
(Hedysarum Alpinum)

like plant found in the northern
and on the tundra. The flowers
pink-purplish and look somewhat
blossoms. The roots are quite
substantial and can be eaten raw or

A vetch-
forests
are
like pe
boiled.

Woolly Lousewort
(Pedicularis Lanata)

The roots and young flowering stems may be eaten raw or boiled. A common perennial of the arctic tundra, up to 20 cm in height.

Silver-weed
(Potentilla Anserina)

Has widespread distribution. The roots are a perfectly acceptable food when boiled or roasted.

Lamb's Quarters or Pigweed
(Chenopodium Album)

Common throughout North America. The young leaves of this plant make an excellent green. Boil like spinach.

Stinging Nettle
(Urtica Dioica)

The leaves of the young plant make an excellent green. The bristled leaves and stems of the mature nettle produce a rash upon contact with the skin. Crushed dock leaves make an effective remedy for nettle stings.

Lady's Slipper
(Cypripedium Acaule)

The roots may be boiled or eaten raw.

Labrador Tea
(Ledum Groenlandicum)

Tiger-lily or Turk's Cap Lily
(Lilium Superbum)
Has a brown spotted, orange flower.
After peeling, the roots may be boiled
or eaten raw.

Pond Lily
(Nuphar Species)

Widely distributed, the roots may be
eaten if peeled and boiled. The seed
pods are best when newly formed. Boil
in two changes of water.

Widely distributed. Leaves make an
aromatic beverage somewhat like Chin-
ese tea. Dry the leaves over fire in a
pan until they become crumbly. The
infusion is prepared by pouring boiling
water over the leaves. The resultant brew
is high in vitamin "C". ¥

B-GA-217-001/PT-001

CARE OF
EQUIPMENT

CARE

AND

USE OF

EQUIPMENT

AND

CLOTHING

INTRODUCTION

In the survivor's battle of man against nature, the odds greatly favor the man who is able to utilize his available equipment to the full and knows how to care for it.

It has been found that some persons have failed to survive, even under reasonably good conditions, not from lack of equipment but from failure to care for it and use it to the best advantage. Here are some suggestions for the proper care and use of the various pieces of equipment found in your survival kits, together with some suggestions for the fabrication of further survival aids.

Axe

This is one of the most important pieces of survival equipment, and also one of the most abused. Properly used it can simplify your survival problem, but misused, it can become a means of crippling yourself to such an extent that survival becomes impossible.

If your axe does not have a sheath, make one from any suitable material available, and keep the axe in its sheath until it is required for use. Before using, make the following checks.

- (a) Check the head for tightness of the handle. If it is loose either drive the wedge further home or make a new wedge using hardwood. Soaking the head is another method but it is not recommended for winter time, as ice may form on the handle and inside the head, allowing the head to slide off and cause possible injury. To drive the handle further into the head, strike the end of the handle not the head of the axe.
- (b) Check for sharpness. A dull axe can be dangerous for two reasons, First it will not bite properly and will tend to glance off the wood being cut. Secondly, when blunt it is necessary to use more force, which usually means a sacrifice of control.

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B-GA-217-001/PT-001

(c) Check that the handle is not cracked or split. A serious cut or clean the head carefully, replace the sliver inight be received. position. It is permissible to store by When carrying an axe, be sure that the sharp edge is held away green wood should never be used for from the body. In the event of a fall, there will then be less chance , of injury.

When felling a tree, these procedures should be followed.

- (a) Before beginning, clear the tree of lower limbs, and remove the underbrush from around the bottom of the tree. This is to ensure that the axe is not deflected during the swing.
- (b) Check your distance from the tree to avoid underreaching or overreaching. Overreaching can result in breaking the axe handle:

aan

underreaching in a cut foot.

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(c) Take up a comfortable stance, making sure that both feet are , " firmly set.

(d) The first cut should be made on the side of the tree facing the Vy direction of the desired fall, often decided by the "lean" of

eZ

the tree. This cut should be not more than half-way through the tree. The back cut should be commenced slightly above and opposite the first cut.

(e) It is safest to cut the tree off not over 30 cm above the ground. Always keep the axe handle low, ic, patallel to the ground Always stick your axe

axes

where the blade strikes the cut. When using short handled or hand axes bend fully at the hips or kneel on one knee. in a log or stump

When splitting wood, do not lay the piece to be split on the ground, but support it as illustrated. This not only prevents the axe from chopping injury to into the ground and becoming blunted, but may also prevent the legs or feet.

should be kept sharp and carried in a sheath immediately after use. Always position back of the hip, since with the knife possible that a fall could drive the knife into the attaching a cord from the handle of the knife throw your knife. It is ineffective when damaged or lost.

experience in the handling of firearms proficiency in their use, a rifle or shotgun Practice firing on a range is the best method weapons used in the service, and facilities practice. It is recommended that all aircrew and become as proficient as possible firearms.

When finished with the axe, sheath, and store in an upright sticking it in a dry stump, but this purpose.

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Knife

As with the axe, the knife stout sheath. Return it to the the sheath on your belt towards th in a forward position it is groin.

Guard against loss by to your belt or belt loop. Never so misused and it will probably be

Firearms

To the survivor who has had and has acquired a degree of ean be an invaluable asset. of getting acquainted with the exist at most units for such take advantage of these facilities in the handling and firing of

Rules for the proper care of your weapons are as follows.

(a) In cold weather, store your firearms in a sheltered cold spot. course essential and the clothes on your back may be

Avoid moving between the warm shelter and cold outdoors as have till you are picked up.

eventually, because of the condensation of moisture on the clothing wisely, making every effort to keep it clean and metal parts, the protective blueing will break down and allow If your clothes become wet, dry them as soon as possible.

the metal to rust. Also remove grease and oil from the weapon particularly should be kept dry. These will usually get and replace them with non-freezing oil or graphite. If neither day's wearing. Unless too damp they can be dried quite is available, the moving parts can be rubbed with a pencil or placing them in your sleeping bag before retiring and the gun can be used without lubrication. for the night. If a hole is worn in a sock, turn the

prevent a blister on the heel.

(b) Keep the barrel clean at all times. A plugged barrel may cause

dress lightly when exerting yourself, and have extra

a firearm to explode, with serious injury to the user.

put on when sitting around idle,

(c) Always prove your weapon when storing it or picking it up. in a parka, it is wise to drop the hood and allow

"Empty" guns have caused many fatalities. around your body to escape. The hood can be raised again

(d) Store guns and ammunition in a safe dry place. completed.

Arctic, damp clothing can be left to freeze and the moisture Snow Saw-Knife when frozen.

In arctic survival, the snow saw-knife is clothing may be dried by the fire. Take the following tools in your possession. Guard against loss by snow when not in use. At night take it into the place clothing nearer to the fire than you it may be necessary to cut your way out in the your hand.

ice or frozen meat with it; under extremely cold conditions this can

Never leave clothing by an untended fire. chip or shatter the blade.

drying leather foot-wear, turn it and work the leather Ice Chisel

periodically to keep it pliable. When almost completely dry, apply a good coating of dubbin or fat, working the grease well

As with all edged tools, it should be kept sharp. The sectional handle the seams and pores of the leather. Do not heat leather. should be checked for rigidity before use, and the joints shimmed where necessary, If a cord loop is not supplied, make one and attach it to the Equipment

end of the handle, looping it over your wrist when using the chisel. This will guard against loss when the chisel breaks through the ice. contain many small items which can easily become

after. The following rules apply to such pieces of gear Sleeping Bag

usually carried on the person but are used by everyone

In summer, keep it dry, air it when weather conditions permit, and shake it daily. Roll up lightly when not in use.

designated place for the equipment, and return it after

In winter, keep it free of snow. Open it completely. every morning

Have this location well marked, and make everyone in the and gently beat off any frost which may have formed during the night. aware of its existence.

Roll the bag up lightly and cover it to protect it from the snow when lay equipment down on snow, spruce boughs, or ground. not in use. Air your sleeping bag as often as weather will permit, shaking in your pocket or bang it in a conspicuous place.

it gently to restore resiliency to the feathers. If using wood fires for your equipment in an accessible place, so that you can heat, guard against spark damage. If a hole should occur in your bag such items as your signal flares, at a moment's notice. repair it immediately. Considerable loss of insulation will result if a hole items, such as compass, match container, etc., should be is not repaired when first noticed.

to the person while travelling to avoid loss.

Clothing Rules

Good hygiene is c

the only ones you

Use your

dry at all times.

Socks and mitts

damp during a

effectively by

leaving them ther

sock over to

In winter,

clothing handy to

When workin

the warm air

when work is

In the

beaten out of it

In the bush

precautions.

(a) Never

hold

(b)

(c) When

int

Miscellaneous

Your kit wil

lost if not look

which are not

in the party.

(a) Have a

use.

party

(b) Never

Put i

(c) Locat

teach

(d) Small

tied

4Mher

Generat

It is even more important to keep your clothing clean and in good
repair while on survival than it is during your everyday life in the city.
rope or twine unless absolutely necessary, as you may Do not cut
its original length. need it later in

Make sure your cooking is done on level solid ground to guard against tipping.

Fill your Primus stove, lantern, etc., away from your shelter, and over some utensil that will catch spills, thus reducing waste and eliminating the fire hazard.

meat and fish; and

Make a habit of tying knots which can easily be undone, such as the bowline, clove hitch, etc., particularly in cold weather.

Do not throw away scraps of cloth, bits of twine, metal, and so on. a woven case and seven twisted threads, each

You may find a use for them later.

smaller threads. The complete line will lift about

Keep your cooking and eating utensils clean. Any abrasive material, used for:

such as sand or wood ash or a combination of both, makes a good substitute for soap.

thread, waxed;

During the arctic spring (April to June) and in winter bush snow from threads;

conditions, sun-glasses, improvised or otherwise, should be used continuously. As long as the sun is up, especially on slightly overcast days sewing thread and sutures;

there is a danger of snow blindness. If your glasses become steamed up lashings; and

while working, do not discard them but try to provide more ventilation. webbing.

Snow blindness can be serious.

provide:

Improvised Equipment

mitten, and headgear; and

There will be a number of tools or appliances which would be most useful, but which are not contained in your survival kit. With the use holsters.

of salvaged material such as metal wire from the aircraft, fabric, cord, and webbing from the parachute and harness, and the natural materials around you, you are limited only by your ability and ingenuity in the

Item

production of gadgets to make yourself more comfortable.

Fish spear

Here are a few ideas which have been developed in the past.

Fish hooks

(a) needles from bone or metal parts of the aircraft;

Belts, pack straps, and tump-

(b) fish lures from various combinations of wood, metal, and brightly lines coloured cloth;

(chest packs) Fair quality knives

(c) snow knives from wood, metal, or bone;

pack bands Catapults

(d) water bottles from bark, the internal organs and skins of animals, and waterproof materials from the aircraft;

(e) good whisk brooms from wings of birds;

(f) eating utensils from bone, wood, or metal; and

(g) eye snow shields from wood or from cloth.

The bow drill principle can be used for drilling holes in objects.

- (c) clothing;
- (d) sleeping robes
- (e) sails;
- {f} wrappers for
- (g) bandages.
- The shroud lines have
- thread consisting of three
- 225 kilograms. They can be
- (a) snares to catch
- (b) fish lines—one
- {c} fish nets—woven
- (d) ropes, lines,
- (e) lifelines and
- (f) snow-shoe
- The pack can
- (a) knapsack;
- (b) foot-wear,
- (c) axe and knife
- Miscellaneous
- (a) Wire frame
- (b) Release Pins
- (c) Harness webbing
- (d) Steel plates
- (e) Bungee cords of

The Parachute

The parachute is probably the most useful single item available to the survivor, if he is ingenious enough to utilize it fully.

The canopy can provide:

(a) shelters;

(b) signal strips;

CHAPTER 11

TRAVEL

INTRODUCTION

A survey of survival incidents both in the Canadian Forces and USAF has indicated that travel is not to be recommended except under conditions when the aircraft has been forced down in an area where survival might prove difficult or dangerous. Under these circumstances, travel should be only as far as is necessary to find a good safe dry location for your camp.

Each survival incident, however, must be considered with regard to its own specific problems, and the decision as to whether or not to travel must be made by you, the survivor, and made quickly. If you do decide to travel, you must travel while you still have strength.

There are five basic requirements for travel. If any one of these cannot be fulfilled to your specific situation, then don't travel.

- (a) Know where you are and where you are going. If you do not know where you have landed, you can rarely plan a route to safety. Stay put!
- (b) Have a means of setting and maintaining direction. If you have a hand compass and know how to use it, you should be able to maintain a planned course. If you have not, then you will have to use the Big Dipper and pole-star, or the watch-and-sun method, which will be described later. If you have no sure means of determining and maintaining a heading—sit tight!
- (ce) Most people are inclined to over-estimate their physical capabilities. Even if you are athletically inclined, able to walk eight or ten km round a golf course in balmy summer weather, it is no indication of how far you are likely to travel through boggy country or hip deep snow. Be very very careful when trying to estimate your physical stamina, and if in doubt—don't start out! Clothes make the man, This is particularly true in survival when the proper clothing can mean the difference between life and death. Proper clothing is important for all seasons of the year, as it affords protection not only against the elements but also against the almost inconceivable torture from bites by the hordes of insects

that abound in Canada's summer northland. Adequate foot-wear is Ridges offer drier, more insect-free, travel than bottom land. There perhaps the most important item of clothing. Wet socks can cause will usually be less underbrush and as a result it will be easier to see and grave discomfort and may completely incapacitate a man. Wind- be seen.

proof clothing is a necessity in cold weather and should be worn Larger river crossing should be attempted only when absolutely necessary over an insulating type of underclothing such as woollen under-sary. If the water is deep remove all clothing, placing it in a bundle, and wear. Unless your clothing is sufficient to protect you from any replace your boots without socks. Boots give better footing on the river conditions which you may encounter-sit and wait!

bottom and prevent injury to the feet during crossing. If forced to swim in (e) Food, fuel, shelter and signals, must be considered in relation to fast flowing rivers, start up-stream from your proposed landing place and the type of country and the season. The emergency rations let the current drift you down to it. When fording a fast shallow stream use normally carried in aircraft will be of little help as they are not pole to help maintain footing, by placing the butt-end down on the up-designed for active survival. They provide much less than the stream side.

caloric intake required for travel and must be supplemented. Fuel Decide whether to cross or go around each lake. If it is decided to is no problem in bush country, but travel on the tundra, especially cross, use a raft or floatation gear to assist. Swimming cold waters can be in winter, will mean transporting sufficient fuel. A small mis-risky, Play it safe.

calculation could be fatal. It is advisable to carry sufficient Deadfall and swamps should be avoided. Deadfall can be dangerous, materials to construct a basic shelter. A piece of parachute cloth because of the ever present danger of slipping and injury. Swamps can also or tarpaulin is all that is required. A sleeping bag should be be dangerous, but their main problem is the steady sapping of strength be-carried during all seasons. Signalling aids are a necessity for the survivor and constitute a definite requirement while travelling. cause of difficult walking conditions. Go around them.

The heliograph mirror and pyrotechnics should be carried in your Mountain areas have their own particular problems. Watch for over-pockets or on the top of your pack, where they can be put to use head threats, shale slides, etc. When crossing shale slopes, it is advisable to at a moment's notice.

rope the party together, and send one man at a time across the slope, using the remainder of the party as an anchor against a possible slide. In early

The following equipment is suggested for most situations; sleeping bag, spring, cross mountain streams in early morning to avoid the greatest fabric for shelter, waterproof match container (full), food, candles, cooking volume of water, which occurs when the sun starts melting the snow. When utensils, axe, knife, gun and ammunition, fishing gear, maps, compass, extra «crossing snow slopes in summer, it is less dangerous to cross them early in clothing, first aid kit, sewing kit, sun-glasses, signal mirror, and pyrotechnics. the morning when they will have a hard crust.

These are all small items, but considering them together in relation to (c), it is enough to make you pause and reconsider. Remember, if it looks like winter

a rough go you will be far better off to remain where you are.

Game trails, especially if heavily used, will save walking through deep BUSH TRAVEL

'now, but you must avoid being led off your general heading.

Summer

Streams and rivers will provide your best method of travel, being the highways of the Canadian north. There are however, dangers in winter river Bush travel in summer is relatively easy, if the following rules are

' «el which must be carefully watched for and avoided. In certain places followed.

+ ng the river, weak ice will be found, and it is best to know in advance Before any bush trip, climb a high hill or large tree to orient yourself @here to look for it.

with the surrounding area and possibly discover human habitation.

Game trails provide an easy path through bush country. The main game

(a) Stay away from rocks and other protrusions, since ice formation in y these localities will have been retarded by eddies.

trails follow the ridges and river flats and are connected by a network of trails. The danger in following these trails lies in the fact that, unless you

(b) Walk on the inside of curves, since on the outside of curves the keep a careful check on direction, you may wander off your heading.

river current has an eroding effect on the under side of the ice

Streams may be followed to larger rivers or lakes, along the shores of surface.

which you are mostly likely to find habitation. Generally, it is better to

(c) Take to the bank or walk on the opposite side of the river at the follow the drainage pattern than to cross it.

junction of two rivers, because the currents from both rivers

Rivers may be followed along the bank, but the winding nature of

hold up the formation of the ice through turbulence.
rivers usually means travelling about four times as far to get from A to B
(6) Stay on clear ice when possible since a deep layer of snow will in-
as opposed to ridge travel. Unless the water-ways in the survival area are
sulate and retard freezing, and erosion by the river may leave only
well known to the survivors, raft building is not recommended.
a snow bridge.

(ce) Carry a pole for testing ice and for use in supporting your weight
SEA ICE TRAVEL
if you break through.

(f) Be prepared to get rid of your pack if you should fall through
Food in the form of seal, fox, and polar bear is more readily obtained
the ice.

sea ice than on barren land. Unless serious injury prevents travel,

(g) Before beginning any trip on ice be certain that a good water-
split up the party as there is generally nothing to be gained by anyone
proof fire starting kit is immediately available and will not be
remains behind. All should travel at once.
lost.

problems of navigation are identical with those on the barren lands

(h) You may encounter overflow under the snow. Wet feet will freeze
very great exception. The polar ice pack is in constant motion. This
rapidly.

by current and wind. Therefore, determination of direction may

particularly difficult. Also one rarely travels in a straight line, but follows

Ridges may give easier walking conditions as they do not usually have
smooth going and avoids the rough ice. Landmarks in the form of high
the same amount of snow as the valleys.

ridges and hummocks are usable only over short distances, since

Mountain areas, in winter, can be particularly treacherous, with the
that they are located on other floes which are changing position.

possibility of snow slides, uncertain footing and sudden storms. Snow

this the fact that the magnetic compass is very unreliable in high
slides will occur from natural causes, but care should be taken to avoid
and the necessity for constant directional checks on the sun and
causing them through carelessness.
becomes obvious.

Deadfall is even more dangerous in winter than in summer since a lot
ice in the very high latitudes is comparatively solid in winter. As
of it will be covered by snow, making walking conditions extremely treach-
returns the ice recedes and there is open water along the entire
erous.

coast. Along the north coast, ice lies off shores, and with strong north

winds floes are often driven ashore. Riding one of these floes is

definitely a Jast resort procedure, since there is no guarantee that the wind
Snow-shoes and skis are not essential on hard snow. On the Arctic

continue until the floe grounds. Ice floes in the fall are Jess dangerous ,
Islands and barrens east of the 142nd meridian, walking conditions are

will usually not travel too far before they freeze in again.

normally good in winter. In some localities frequent gales are encountered.

The summer ice is covered with lakes and water soaked snow, which
There is no shelter except that provided by scattered high banks and willow

drain off as the ice develops holes and cracks. There is practically
thickets around lakes and along stream beds. Game is very scarce and fires
surface anywhere. Fogs abound and misting rain is frequent. All

cannot be maintained for long on the fuel generally to be found in the dead
been said about winter travel applies to summer travel. Survivors

of winter. The survivor cannot afford to follow the streams which, because
leave the ice and get to land if this is possible. Travel should other- °

of their winding nature, double and quadruple the distance to be covered.
restricted to a minimum since it will avail litde and it is dangerous

The compass js not reliable and landmarks are few and far between. One
exhausting.

man will have difficulty steering a straight course by himself. Two can do a
Cracks in stationary ice, €.g., on sounds or bays, do not pose any great

little better but three are required to navigate when visibility is low. It is
during the winter, and in the spring and summer they are readily

recommended that any extended travel over barren Jand or sea ice be done
have to be circumnavigated via their narrow end. Cracks in moving

by a party of at least three for this reason.

be opened and closed quite suddenly by wind and tide pressures '

The spring break-up, summer, and the fall freeze present far greater

should be on the alert for them at all times, using both eyes and
travel difficulties than does the winter season. Equipment must be racked

on the back. The masses of soggy vegetation on the tundra cause the traveller
to slip and slide. Lake systems must be either crossed or circumnavigated.

icebergs frozen in the ice are likely to have open water in their

Care must be taken in crossing sand-bars and mud flats at the mouths and

This is because of the force exerted by the current on the greater!
junctions of rivers, and lake and lagoon outlets. Quicksands or bottomless

the berg below the surface. Icebergs driven by the currents have

muck may trap you. If a life raft is available, it is preferable to float

known to crash through ice at least one metre thick. Towering ices

down a river rather than attempt to travel across country. The months of

always a danger in open water. The area below the surface

July and August are about the best summer months for cross country travel. faster than that in the air. When the berg's equilibrium is upset, It Because of the prevalence of fish in all streams or lakes, a fish net is one over, and the adjoining area is no place for man or beast. The of the best pieces of equipment the traveller can carry. A rifle may provide tidal waves throw the surrounding small ice pieces in all dircce game for a number of meals. Therefore, stay away from pinnacled bergs. Seek out low, flat

bergs for shelter at sea.

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topples
resultin
tions,
topped

TRAVEL HINTS-SUMMARY

General

larger rivers in the interior of Alaska and Canada are main lines of summer and winter travel.

Take it easy. Conserve energy. If you are tired, stop and make camp. Use topography to the best advantage.

drainage

Carry adequate but only necessary equipment. Don't overload. Keep lines not across them, particularly in summer.

your equipment in good condition; guard against losses; and protect food Avoid climbing if possible.

from prowling animals.

Be prepared to signal passing planes. Messages

Don't travel alone if it can be avoided. Mark the trail and leave mes- These

sages wherever possible. left at the

crash scene. A message should also be left at any point at which a

To follow a straight line choose two easily visible points ahead in the following

change in plans is put into effect. The message should contain information:

back occasionally will also help to keep you in line with your departure (a)

date of leaving the crash scene; (b)

point. (c)

destination and route; (d)

Plot your course on a map. (e)

estimated length of the journey; (f)

Improvise travel aids such as snow-shoes, sleds, rafts, and packs.

number in the party;

Keep a log-book of your travels.

physical condition of the party; and

Take care of your feet.

any other pertinent data regarding supplies, ett.

certain the message is left in a readily accessible place. Lacking Make

In Summer @ pen or

pencil, messages may be written with charcoal.

Dense vegetation, rough terrain, insects, soft ground, swamps, lakes, Uses of

Compass

and large rivers are obstacles to foot travel.

Prospector is the compass packed in Canadian The Silv

Cross glacier-fed streams in early morning to avoid the maximum Force

kits at the present time. It may be used as a protractor survival

a volume. or a

Travel on ridges and game trails is the easiest method-maintain compass.

the Compass as a Protractor Using

constant direction checks.

straight line along your desired direction of travel indicating Draw a

In Winter A as the

present location and B as the intended destination. This line is your require

Soft deep snow, dangerous river ice, severe weather and a scarcity

track. Plac

of natural food are major obstacles in winter travel.

the compass along this track as shown in Figure 1.

Don't travel in a blizzard or during extremely cold weather. Hole up

Op Bee

and save your strength.

When travelling on frozen rivers be wary of thin ice, overflows, ice.

Lyi PR OK

heavy snow, and air pockets. Use a pole or ice chisel for testing

Rope together when forced to cross dangerous ice.

If you do break through ice get into shelter and start a fire.

Guard against frost-bite.

: LEE

In Mountains

(isa?

J} ty te BegIndex Pointer

In this region, routes should be chosen to lead you:

f TAD .. F

! Pins ge

{a} out of a dangerous location to a more sheltered area;

(b) to the seacoast;

"a

Required Track
(c) to a major river; or

AA

(d)} to some known point of habitation.

Travel on a trail, down a river, or along a seacoast will usually lead
to a cache, a cabin, or a settlement.

Figure 1

IN.

Rotate the compass housing until the compass meridian lines on the transparent compass face are parallel to the meridian lines of the map and north (N) points to true north on the map as in Figure 1.

heading established and set on the compass it is now

The compass face is marked off around the perimeter in 2 degree intervals represented by a short line, a larger line every 10 degrees, and held in your hand turn yourself around until the Azabic numbers every 20 degrees. The number 2 represents 020 degrees, and seeking pole is pointing to north (N) on the compass number 4 represents 040 degrees, number 26 represents 260 degrees, and needle is parallel to the two luminous lines, so on with north (N) representing 360 degrees.

Now read off your required track opposite the index pointer. In Figure 1, your required track would be approximately 040 true.

Variation

You now have the required track true. In order to travel along this required track, you must allow for local magnetic variation. In other words, you must compensate for the angle between true north and Magnetic north. Variation will be either easterly or westerly and is always indicated on topographic maps.

A good rule for applying variation is:

Variation east-magnetic least; and

Variation west-magnetic best.

EXAMPLE. Figure 1 indicates that your required track from A to B is 040 degrees true. Let us assume that the local variation is 20 degrees east. Therefore, according to our rule, we would subtract 20 degrees easterly variation from our required track true of 040 degrees. This gives us a magnetic heading of 020 degrees which is set on the compass opposite the index pointer,

Sighting the Compass

Sighting the Compass

pointing along the required track and a sighting

illustrated in Figure 3.

is read between the two luminous compass orient-

reflection in the sighting mirror. In using the sighting mirror,

the needle is oriented between the two parallel lines,

to one line than it is to the other. This would give the

not exactly in the centre (Figure 4). However, this is

mirror, slanted as it is at a 41 degree angle, influences

to the laws of parallax. In using the sighting mirror,

needle with the orienting line which appears nearest

will follow. Be sure to level the compass while

mirror.

How to Use the Compass

With the magnetic

ready for travel.

With the compass

"Red End" or north

face and the magnetic

Figure 3

The compass is now

tine may be obtained as

The magnetic needle

bmj lines by its

you can see that, when

-rpoars to be closer

wry cession that it is

fully explained. The

the reflection according

sherefore, parallel the

tM, and accurate results

talag the sighting

Right

The magnetic needle is closer to one of the lines than it is to the other. distance would be an asset at times,

and explore its potential;

in order to resume track on the opposite

the ground to a 1: 1,000,000 map is not

distances on the ground to scaled distances

simplify the task of interpreting the map.

estimating distance is the tally and pace system. It is

Sighting Mirror
adopted for most military drill work.

short pace for most people. It is a pace which

=75cm

time =150cm=1.5m

Correct Position of Needle

= 102 m = 1 tally (tie a knot in a

Figure 4
string for each tally)

. = 840m=1km

for you to learn all the symbols shown

know the ones which indicate difficult

locations. Generally speaking these are as

Sighting Mirror

The dotted lines indicate that the exact
marsh or swamp is not known.

+ JE: Wrong Position of Needle

ug. le. | ©

at ak ee - = .

= ae ~

= "iu

Wrong

The magnetic needle appears to be centred between the two luminous lines, but is not parallel to them.

Choose a landmark to which the sighting line points (rock, tree, etc). Walk directly to that landmark. When you have arrived at your landmark, sight again, choose a new landmark or target, and travel to it as before. By moving from new landmark to new landmark, you will describe a straight path through the country-side to your destination.

Swamp or Marsh

Measurement of Distance

A simple means of determining

when walking, e.g.:

(a) to survey the crash area

(b) to make a dogleg detour

side of an obstacle; and

(c) to orient oneself on

easy-relating actual

on the map will

One method of

based on the 75 cm pace which is

This is neither a long nor a

you probably use every day.

1 pace

1 double pace (i.e., each

the same foot touches the ground)

66 double paces

16 tallies

Map Reading

While it may not be possible

on various maps, it is as well to

or impossible travel or aid

follows:

{2} Swamps or Marshes.
shore line of the

#4

vt

7 ary

aa

B'S

(b) Undulating Country. The height of the ground is indicated by series of light lines drawn across a continuous line joining all points of equal height. The variation which the rapids occur. They are in height will differ depending on the scale of the map and rivers.
the terrain which it covers.
This shows that a steep slope is represented by contour lines close together while a shallow slope has contour lines far apart.

as a light double line drawn
the word "falls" written alongside,
in metres.

(c) Rapids. These are shown as a
the river covering the area in
usually marked only on navigabl

Rapids

(d) Falls. These are generally show
directly across a river, with
and a figure indicating the dro

(e) Trails. These are usually shown as dotted lines, either single sun and your watch, is a little more or double. A single line usually denotes a pack trail, while accurate. First, by orienting your double lines usually indicate a trail used by wagons or tractors. directly at the sun. Then by bisecting

hand and twelve o'clock you have an

(f) Cabins. Ranger stations, etc., are marked as black squares with "Cabin", "Mission", or "Lookout Tower" written alongside. and south. For example, at 8 am it Trading posts are shown as black squares with TR written alongside.

It is best to remember that the shortest distance between two points

is not necessarily the quickest route. Pick your route carefully to give you easy walking conditions, even if it does add a few kilometres to the trip. It will pay off in the long run,

Finding Direction Without a Compass

Three methods of finding North are discussed here.

(a) The first, by using the pole-star, is the easiest. Having found the pole-star, simply face it and you are facing North. To find the pole-star, the Big Dipper or Plough is used.

Determine a North-South Line

the sun is up all day, any con-

12 noon could cause an 180

the example shown above, had the

rather than 8 am, the direction would

watch is on local or standard time.

north is to place a long stick in the

progresses place shorter twigs in the ground

shortest shadow will indicate north,

short twig to another will indicate the

Blazing

purpose on a survival trek. It will not

party, but will also help the survivor to

should he change his mind about

The Big Dipper and Pole-Star
168

(b) The second method, using the complicated, but reasonably watch, point the hour hand the angle between the hour imaginary line running north would appear thus:

Q@Q%
R

Using the Sun and a Watch to

Remember, in the Arctic, when

fusion between 12 midnight and

degree error in direction. In

observation been at 8 pm

be reversed. Make sure your

{c) The third method of finding ground, and as the sun at the end of the shadows. The or a line drawn from one east-west line.

Blazing serves more than one emly be a guide to a ground search @mi his way back to the crash scene Wavelling.

*

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In the bush, blazes should be placed on both sides of trees.
available, long grass may

In grassy clearings, where trees are not
be tied together in bunches.

Direction amu

.

Right Turn

Message

Tree Blazing

In short growth, saplings may be snapped.

winter, where nothing else is available,

In the Arctic, or bush country in
'ow blocks may be used to mark 2 trail.

cyUeto

oa

aT

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In summer use rocks or sod blocks.

SEA SURVIVAL

CHAPTER

12

SEA SURVIVAL

To have to survive a bail out or ditching at sea is probably the most formidable experience one can expect to be faced with. Man's natural environment is land and every action he must perform in a sea survival situation must be learned either through study or practice.

When one considers that approximately 71 per cent of the earth's surface is covered with salt water, aircrew can expect to fly over salt water for a large percentage of their service career. In the event of an emergency, when flying over the sea, it will probably not be possible to reach shore prior to having to bail out or ditch the aircraft.

Survival at sea is undoubtedly the most severe survival situation that man can be exposed to. To survive for any reasonable length of time, even in tropical waters, one must have protection from exposure, water, food, and protection from various dangers of the sea. In short, one must have survival equipment in the form of flotation equipment, 2 means of protecting himself from the elements and other danger, as well as drinking water, or a means of procuring it if survival is going to be successful. In addition, a survivor or survivors at sea must have artificial methods of attracting the attention of search aircraft or vessels, if rescue is expected. A life raft at sea is very difficult to spot from the air particularly if there are waves and/or if the area one may be down in has not been pinpointed. For aircrew the life preserver, life rafts, and ancillary equipment available provide these life support aids. However, having the equipment alone is not sufficient—one must know how to use the equipment properly, how to take care of it and how to get the most out of it if one is to survive until rescued.

Preparation for Sea Survival, as for any survival, must start long before the emergency arises. This includes being aware of the possibility of an emergency, knowing what survival equipment is available, its location in the aircraft, and above all, knowing your bail out procedures and/or ditching drill.

'The Sea, like the Arctic and desert is unforgiving; any serious mistake will probably be your last.

Immediate Actions After Ditching

ing one shroud line to the edge of the canopy. Clear all shroud lines
The life raft should be boarded directly from the aircraft, if
possible. from your legs and other possible areas where they may be entangled and
Step into the raft—do not jump in. If the raft inflates upside down one
swim away from the canopy.

man should enter the sea and right it. If it is necessary for all survivor
If the wind is blowing so strong as to keep the canopy inflated, it will
the

to get into the water before boarding the raft, one man should board
undoubtedly pull you through the water if you have not either released
raft first, then assist any injured personnel aboard. Ditching suits should
your Capewell releases or activated the quick release box and cleared
be worn by all, if available. the

leg straps. If this should happen, to prevent drowning, and if you are not

Roll should be called or a head count taken, before cutting loose from
already on your back; roll over, bend sharply at the waist and spread your
the aircraft to ensure all personnel are accounted for and to make certain
legs. This will put you in a planing position to prevent you from porpois-

all survival equipment is on board the raft.

ing through the water. At this time, take the necessary action to release
The raft should now be cut adrift from the aircraft and paddled free
yourself from the canopy.

before the fuselage sinks. If more than one life raft is used they should
As soon as you are free of the canopy, pull your life raft to you, if it
ately

rendezvous and tie the rafts together with a piece of line approxi-
was inflated in the air, and board it. If it was not inflated in the air, inflate
eight m long. This will prevent the rafts from rubbing together and will
it at this time. Do not hold on to the CO' bottle as you can receive a frost
in

also prevent them from jerking with the wave action which could result
bite from it as the bottle inflates the raft.

damage to the rafts. All equipment should now be lashed to the raft and

If your parachute harness is equipped with Capewell releases, ensure
one occupant of each raft should attach himself to the raft with a length
the covers are closed. Board the raft from the foot or narrow end by grasp-
of line. This is a precaution to prevent the raft from drifting out of reach
ing the handles on either side and pulling it down and under you. Then
in the event it overturns. Check the ballast pockets to ensure they fill
progressively pull and squirm the rest of the way into the raft taking care
with water and stream the sea anchor. This line should be adjusted so that
notto puncture -or tear it. Roll over in the raft and take up a sitting
the sea anchor will remain in the trough of a wave or swell when the raft
position.

is at the crest.

Ensure the ballast bags are filled with water and stream the sea anchor

Administer first aid as required as soon as possible.

(it is normally stowed in the forward ballast bag). If necessary adjust the

J€ weather is adverse, erect the canopy or if your raft has a self
sea anchor line so that it remains in the trough of the wave when the raft
inflating canopy close the side curtains. Check the raft for damage, top
is at the crest.

up the buoyancy chambers if not firm enough. These should be firm but

, Bale out the raft, wring as much water out of your clothing as possible,
not drum tight.

bring your equipment pack aboard, and finish bailing using the sponge until

Bale out the water, finishing off with the sponge. With patience
it is

the raft floor is virtually dry. If weather is inclement and you continue to
to eliminate ail water from the raft in this manner. If

clothing is

possible

ship water pull the spray shield over your head and close it completely.

wet, it should be removed and wrung as dry as possible, then put back on

If the raft is too soft, top it with air using the hand bellows. It

prior to final sponging of the raft floor.

should not be drum tight as it will ride too light in the water and be very

You should now prepare all of your signalling devices and have them
wusceptible to capsizing. It is better to have it soft enough to bend with
available for immediate use. Use your emergency radio in accordance with
the waves.

instructions and at the prescribed times.

Keep the raft and all equipment tied to your person with a length of

Activate the desalter kits and solar stills, if available, in accordance
Vine to prevent loss in the event of capsizing or accidentally dropping
with instructions.

equipment overboard.

A definite plan of action should be taken. Normally, the aircraft
captain takes charge but he may elect to delegate someone more

experi-

enced. Good leadership is of utmost importance. Duties should be allocated

Qe Raft Environment

to all uninjured personnel.

If the raft becomes damaged it should be patched immediately with
Immediate Actions Upon Water Entry after Bail out
the patches provided. Improvised patches can be effected by using adhesive
As you are descending in your parachute you should activate your
(ape from the medical kit or other adhesive tape that may be available
life preserver and take the necessary action to deploy your seat pack, if
(vom other parts of kits even though it has been previously used. Tape will
the type of equipment you have dictates this procedure.
kk best if the raft surface is dry.
'As you enter the water you should release yourself from the parachute
Air expansion caused from direct sun on the raft or air contraction
by pressing the quick release box (after unlocking it) and clear
the leg from cold air at night might make it necessary to bleed off or top up air to
straps. In the case of parachutes employing the Capewell releases, activate
Maintain the correct firmness of the raft.
the releases on water entry and remain in the harness.
Keep your mind occupied with the task at hand and relax. In a group,
If the canopy settles over you in the water—stay calm, float in your
@ull together as a team, ensure everyone has a job to do. This will help
life preserver, and without kicking your feet pull the canopy off by follow-
(une to pass and will also help to minimize the possibility of sea sickness.

Medical Care and Health Protection
Salt Water Sores

er 4. Place any casualties on
Treat any injuries as detailed in Chapt
a. these are caused
these a 1
by prolonged exposure to salt water. Pre i
and make them as comfortable as possible. Insulation
lies in keeping all clothing as dry as possible; and vention
the floor of the raft nt to help keep the patient
between the floor of the raft and
will be required
b. the sores should be cleaned and an antiseptic cream applied. i If :
him as warm and dry as possible.
they are large they should be covered with a dressing
there are a number of factors
In addition to treatment of injuries,
Squeeze the sores. eoeies Wl
that can affect your health and well d that are peculiar to a Sea
tion. These are chief ly cause by shortage of fresh water
Survival situa
Parched Lips and Cracked Skin
water.
and exposure to the weather and salt
Prevention and treatment are the same. Apply sun screen cream or
Exposure to Wind and Sun
vaseline and do not lick your lips. Cover to i
by sun and wind, if possible. to the sun and wind. In hot
retent ohee 'Seying: Out
a. do not expose yourself needlessly
is better to keep a light layer of clothing on than to
climates it
ely hot, a cooling effect can be obtained
shed all clothing. If extrem
Constipation and Difficult Urination
wringi ng it out, and putting it back
by dipping clothing in the sea,
anti-sunburn cream on areas of skin that must be
Neither of these afflictions should be of concern as they are normal,
Neit
on. Use
considering shortage or lack of food, inactivity, and lack of fresh water. °
exposed;
or sunburn is to apply an antiseptic
b. the treatment for windburn
emulsion and cover.
Signals
As in any survival situation
t all signalling
i devices
i should be checked:
Sore Eyes
be prepared as soon as possible after boarding the raft. Each item eid
reflection off the water will
a. wear sun glasses or eye shields as the
ee ise oe person or the raft so they are not lost in the event of
sun's rays causing sore eyes, Very quickly. Remember
«aps 'ing.
ig. They should be kept as dry as possiblei and ready for immediate
i
intensify the
day this precaution should be taken; and
that even on an overcast
them. Apply an antiseptic
b. if eyes do become sore, do not rub
Use
J the : emergency radios as directed
Us bearing in mind
i to cons
bandage lightly.
batteries until you are confident search aircraft will be looking for you 4
cream to the eyelids and
cold climates remember to keep the batteries warm. For Aurora aircraft
Seasickness
use the Gibson Girl, as detailed in Chapter 6.
various chores will help to
| Do not ignite pyrotechnics until an aircraft or ship is seen. Wh
a. keeping your mind occupied with
effort should be made to remain
'aning pyros keep them clear of the body and don't allow any of the
prevent seasickness. Every
burning residue to fall on or in the raft.

relaxed.

and

try Use the heliograph mirror any y time
 ti! the sun isi shining.
 ini

b. if you become seasick do not eat or drink, lie still, relax
 he bh p Sweep 1 tablets use

horizon with it even if a ship or aircraft cannot be seen or heard, ae
 to keep as warm as possible. If you have anti-seasick
 ney ration can (polished up) can be used as an improvised mirror.
 them.

ut es X in the center of it and aim the same as the heliograph. :

Immersion Foot

n 9 not use J the sea marker dye until an aircraft or ship ship is is si:
 legs to cold water for pro-

ae it as directed on the container. If you are not located oe
 a. this is caused by exposure of feet and
 periods, It is recognized by a red coloring and pain.

we i can be brought aboard and used a second time. Precautions should
 longed

ans isn to Pen i from contaminating your food or water. It can be
 extreme cases, the affected area will swell and
 on the skin;

ow Following this, in
 ¢ outs ide of the buoyancy chamber of the raft, above ¢ the water
 with dark blotches
 blisters will appear interspersed
 by keeping the feet as warm and dry as
 Tight fitting boots

b. prevention is possible
 here 'The signal light should be used at night only. The water activated
 dry will assist.

possible. Keeping the raft floor
 . jery can sometimes be used a second time, if not depleted, by lifting
 should be removed; and
 jammy the water and recapping the holes.

c. treatment consists of moving feet and toes to assist blood circula-
 as dry and warm as

The whistleile should be used t © attract attentioni of personnel in the
 tion, Do not massage them. Keep them
 by wrapping in pieces of dry clothing or dry parachute
 water when originally boarding the raft or in the case of capsizing. It also
 possible and keep the
 cloth. Keep the feet raised clear of any water
 wold be used to attract the attenti tion of passingi surface c!
 body warm.

te-cle well over water. ° raft, 9s sound

ema

Water f. smoking increases thirst so if you must smoke do so during the DO NOT DRINK SEA WATER. evenings or nights; and Fresh drinking water will be your most critical requirement. The body g. to decrease the desire to drink, suck on a button or piece of requires water to keep functioning for an indefinite period. With no water cloth. This increases the saliva in the mouth. to drink one can expect to survive only a few days where the humidity of the air is low and the daytime temperature is in the vicinity of 49°C.

Food

This survival time without water will increase to approximately 10 days The amount of food that one may eat in a sea survival situation is in at lower temperatures, at sea, where the humidity is higher. To direct proportion to the amount of drinking water available. The body appreciably increase survival time, water intake must be increased to requires water for the digestion process as well as for the elimination of approximately 4.5 & quarts daily. There are case histories of personnel surviving waste products, resulting from eating. Food is not that important that one without drinking water at sea for up to 17 days and others who survived on as should be concerned about not eating for several days. The body will little as 57 to 114 g of water daily. However, these cases are exceptional and continue to function for a long time without food as it will draw on body one should attempt to consume as much water as thirst dictates to keep the fats to keep it going. There are two main categories of food in relation- body functioning normally. There is no real advantage to rationing water other ship to the balance between food and water. than to ensure everyone gets his fair share. Drink but do not waste. Injured personnel should have all they want, if available.

a. carbohydrates—such as the emergency ration carried in survival kits. This food requires very little water for digestion. If slightly want, if available. more water is available than is required to maintain a water

Sources of Drinking Water

balance in the system this ration can be eaten as per directions

a. your life raft kit contains de-salter kits and in some instances on the ration tin; and solar stills. Activate these as per instructions on the packets;

b. proteins—such as fish, shellfish, meat, eggs, and certain greens,

b. rain water—use every means available to catch and store rain

This food requires a great deal of water and as a result should water. If salt is dried on the canopy of the raft rinse it in sea not be eaten unless your supply of water is such that you have water prior to catching rain. The small amount of salt contami- more than you need to allay normal thirst. nation will have no ill effect and in fact will make the rain water more palatable;

General Rules

c. dew can be collected on the inside of the raft canopy during the

a. If you have ample water, eat whatever protein foods that are night. This will not produce much water but every bit counts; available first and save your emergency carbohydrate rations for

d. icebergs (in the northern hemisphere) are a source of fresh water the time that water becomes more scarce; but should be approached with caution as there is extreme danger

b, fish will probably be your largest possible source of protein. The in the event they roll over; juice from fish flesh should not be considered a substitute for € old sea ice can be recognized by its clear bluish color and is drinking water as it contains proteins and will require more water virtually salt free; and to digest. All sea birds will also be a possible source of protein

f. some survival kits contain canned water.

food. Sea weeds are edible either raw or cooked but remember all of these require extra water for digestion;

General Rules

c. remember the amount of drinking water you have will deter- Preserving water in the body is equally as important as consuming mine how much, (if any), food can be eaten; and water. Following are important rules:

d. do not eat if you are inclined to be sea sick as it will merely

a. if no water is available—do not eat. The process of digestion, aggravate the situation.

e particularly proteins, requires water to assimilate;

Sebsidizing your Emergency Rations

b. in hot climates reduce the loss of body water through perspira-

tion, as much as possible. Remain inactive. Dip clothing in sea, lf you have ample drinking water you should consider subsidizing wring out and put back on—then use cooling effect of any breeze. your emergency rations from the sea.

If dried salt accumulates on the body it should be brushed off Most fish in the open sea are edible. Those to avoid are:

with a dry cloth. Remain in shade as much as possible;

- =a. fish that are brightly colored;
- c. sleep and rest will minimize loss of body fluids;
- b. fish covered with bristles or spines;
- d. prevent seasickness if possible. Relax and keep your mind on
- c. fish that puff up;
- other chores. Use seasick tablets if available,
- d. fish with parrot like mouths or humanoid teeth.
- e. do not drink alcohol as it dehydrates the body;

Any or all of these can prove to be poisonous. Avoid fish eggs found in clusters or clumps, they will probably be poisonous.

Normal jigging techniques for fishing are recommended or by casting
Landfall
hook out and retrieving hand over hand.
It is generally conceded that one should stream the sea anchor with
Do not handle fish line with bare hands or tie line to the raft.
intent of remaining as close as possible to the scene of the ditching
All sea birds are edible, either cooked or raw. Birds can be caught
bail out. This will normally assist search and rescue to find you.
on a baited fish hook floating on the surface. They can sometimes be
However there may be times when you will have to elect to attempt to
caught as they perch on the raft.
Teach land. A life raft can be sailed up to as much as 10 degrees off wind
direction but your course will be governed mainly by ocean currents
General Dangers of the Sea
wind.
In addition to the danger of capsizing or exposure to the elements
To utilize the ocean current to assist in travelling have all occupants
there are other dangers to guard against.
as low in the raft as possible, soften the raft somewhat, and stream
a. sharks—can be recognized by the fin and tail slicing through the
sea anchor.
surface of the water. They are attracted by light colored objects,
To utilize the wind, if the direction is favourable, top up the raft
blood, vomit, body waste, and garbage. Keep your feet and
it rides higher in the water, have all occupants sit up erect, haul the
hands in the raft and discard garbage etc, in small amounts as
anchor aboard, and empty the ballast pockets. Use anything avail-
far away from the raft as possible, preferably at night. If a
for a sail.
shark approaches the raft attempt to scare him off by making
loud splashing noises on the surface with anything available.
Locating Land
Generally, sharks are only curious unless attracted by the smell
Indications of where land may be, can be determined by:
of something to eat, they will get into a feeding frenzy and
a. cumulus clouds forming in the distance on a clear day, are likely
become very dangerous;
formed over land;
b. barracuda—has a similar appearance to a pike. This fish is
b. birds often fly out to sea in the morning and return to land at
aggressive, and if seen, the same precautions should be taken as
night;
for the shark;
c. wind normally blows towards land during the day and reverses
c. swordfish—have a long bill or sword. Although they are not
in the evening;
normally dangerous they have been known to ram a boat, if
d. in the tropics, often the reflection of coral reefs or lagoons will
attacked or wounded;
be seen on the underside of clouds or in the sky. This will be in
d. killer whale—usually travel in schools. They will be recognized
the form of a greenish tint;
by their sail-like fin cutting the surface of the water as they
e. deep water is dark green or blue, lighter color indicates shallower
swim in a porpoising manner. They are black on the back. They
water and may be an indication land is near; and
will not normally bother a raft, but can overturn a raft if you
f. drifting wood or vegetation often indicates land is near.
are in their path; and
not mistake a mirage for land. By changing your height in the raft
e. moray eels, stingrays, poisonous shellfish, and snakes—these will
mirage will either disappear or change shape.
normally be found in the tropics around coral reefs or near shore.
Precautions should be taken to ensure hands and feet are
Landing
covered, when wading ashore.
If you are sailing or drifting ashore there are precautions to be
Precautions in Dealing with Dangerous Fish
observed:
a. ensure your life vest is worn and fully inflated;
a. keep clothing on and keep a good lookout;
b. wear all clothing including shoes;
b. do not fish if any of the above are in the vicinity;
om Pick the best landing point and attempt to manoeuvre towards it;
c. do not trail hands or feet in the water;
d. trail the sea anchor with as much line as possible to slow down
d. remain in the raft at all times;
your approach;
e. do not throw waste food, body waste, etc. overboard during the
f. try to keep your raft on the seaward side of large waves;
the day;
f. stay in the raft until it is grounded on the beach; and
f. if dangerous fish are about, remain quiet in the raft and the

8. once the raft has grounded get out quickly and beach it.

- g. likelihood of them attacking will be negligible; and
if you are in the water, and dangerous fish approach, you should

General

beat the water with strong regular strokes. If a group of survivors
Remember that no matter how formidable a sea survival situation
are in the water, form a circle facing outwards and beat the
appear you can survive if you possess a strong will to live, keep
water. If there is an injured person keep him inside the circle.
calm, develop a plan of action, and remember your survival training.

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i DESERT
' SURVIVAL

CHAPTER 13

DESERT SURVIVAL

Introduction

When it is realized that there is at least one desert in each continent of the world, it will be appreciated that the problem of desert survival is a very real one for those who fly over these various areas. One fifth of the land surface of the earth is considered desert and is inhabited by approximately 4 per cent of the world population. There are approximately fifty important deserts. The larger of these are well known: Sahara, Libyan, Arabian, Gobi, and Mongolia.

What is the desert like? It's hot and dry, and it's sparsely inhabited and even more sparsely vegetated. Some of it is mountainous, some of it is sandy, and some areas are gravel. It might be defined as an unfavourable site to terminate a flight.

Desert Hazards

The greatest problem in the desert is lack of water. The amount you require depends on the temperature and your activities, clothing, and shelter. The next problem is exposure to sun and heat, one of the desert's greatest hazards. Day-time temperatures can go as high as 50°C. Exposure to such extreme temperatures can cause heat cramps, heat exhaustion, or heat stroke. The desert sun can complicate this with sunburn or sunblindness or both. The scarceness of animal and vegetable life in the desert makes it nearly impossible to rely on replenishing your food supplies from these sources. Another irritating problem is insects. Flies and sand flies can be bothersome, and mosquitoes may carry malaria.

Immediate Actions after Emergency Landing

Water will be your biggest problem if you are down in the desert.

(herefore, on leaving the aircraft when it comes to rest, take all your water if possible and your safety equipment packs. Stay well away from the crashed aircraft until all fire hazards have passed.

Get into the shade as soon as possible, and keep your head and the back of your neck covered. Evaluate the situation calmly and decide on your course of action. Don't rush, take it easy. The two things that require your immediate attention are first aid and shelter.

First Aid

A water still made out of a tin can and a sheet of clear plastic can
Attend promptly to all injuries. Move all injured personnel into the
produce up to 1.7 & of water a day out of hot desert sands. The still is
shade as soon as possible. Follow the established first aid practices.
made by digging a hole, placing a can or other container in the bottom,
Exposure to the desert sun is dangerous. It can cause heat cramps,
and covering the hole with a sheet of plastic. heat

exhaustion, heat stroke, and a serious sunburn.
The centre of the plastic is pushed down to form a
cone over the
Protect the eyes to prevent sunblindness. Symptoms are burning,
can. As air under the plastic gets hot, moisture from the ground evaporates
watering, or inflamed eyes, headaches, and poor vision. Treat sunblindness
to condense on the underside of the plastic, and the drops running down
by protecting the eyes from light and relieving the pain. Place the patient
the plastic will collect in the container.
in a dark shelter or cover the eyes with a light-proof bandage.

A tube can be inserted in the can and run to the surface to enable
drinking from the container without disturbing the still.

Water

Obtaining water is only half the battle. You must
make it last by
Water is the key to survival in the desert. If you are lucky you may
conserving perspiration. The body gets rid of heat through evaporation
have a good supply in your aircraft. If not your chances of finding a good
or perspiration. As the body fluids drop, perspiration is reduced,
supply may be slim.
body temperatures go up, and efficiency is drastically reduced. An increase
If you are near an oasis, your water problem may be solved early.
of six degrees from normal temperature is fatal.
Although it may feel

They are low spots characterized by vegetation.
cooler with your clothes off. you are losing body fluids rapidly.

Being fully
If you are near the coast, fresh water can sometimes be found in
clothed keeps the heat out, and slows evaporation. The
main points to

the dunes above the beach or even under the surface of the beach itself.
remember are drink water and preserve perspiration to avoid dehydrating.
Look in the hollows between dunes for water which may have collected
The water chart at the end of this chapter will provide
an idea of your
there. If you don't find it there, dig down on the leeward side of the
daily water requirement.

first dune or any place where the sand appears moist. The first water you
encounter will probably be fresh. If you get salt water you must not

Signals
drink it.

If inland, look for dry river beds. They may produce water by digging
Your aircraft radio or emergency set will be your best rescue aid.

below the concave bank on the outside of a curve or bend. Allow the water
Ground signals should be laid out to make your position as conspicuous
to collect by scooping small holes,
as possible. Full use should be made of coloured equipment
which will

Another source of water is dew which can be collected early in the
form a contrast with the natural surroundings. Signal
equipment should
morning from rocks and metals. You may sop up the dew with a piece
be kept: ready for immediate use as soon as an aircraft is
heard.

of cloth and then wring it out. Another method of obtaining water is
illustrated below.

Shelter

One of your first requirements will be shelter from the sun and heat.
Natural shelter is limited to the shade of cliffs or the leeside of hills. In
some desert mountains, you may find cavelike protection
under tumbled
cocks broken from cliff sides. Caves and the walls of stream beds provide

source of shade.

If you remain with the aircraft, don't use the inside of it for shelter
in the daytime. The fuselage will be too hot. Use the wing of the aircraft,
with a parachute or any other cloth as an awning. Be sure to

leave approxi-
mately two feet of open space below the canopy for air circulation. Make
sure the aircraft is securely moored and that there is no chance of its
collapsing during high winds or a storm.
A covered trench will also provide shade, but again
there must be
space for air circulation.

In some desert areas the aircraft fuselage may be used

as a shelter
during the desert winter.
while the daytime
temperatures are about 30°C.
Vires
Fires are rarely required
to boil
tea, and for signalling.

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It gets cold during the night
PLANT MATERIAL
Z TROUGH

in the desert except to purify water,

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Fuel is usually extremely rare and aircraft oil and most useful if available. Fire building is not difficult. Stoves out of any metal container or by building stones in a small circle. Fill the that the objective can be reached on the water supply available. Do not container with sand, saturate it with oil or gasoline, and then carefully underestimate the difficulties that will be encountered or overestimate your light with a match. Holes should be made in the container to provide physical condition. ventilation. Never add petroleum fuels to a fire already started or even After it is decided to travel, ensure that you don't overload. Carry smouldering. only the essential items. Your total load should not exceed 16 kg. Water All twigs, leaves, stems, and underground roots may be used for fuel and a shelter will be your main items. Suggested items are: a light material if you are able to locate plant growth, Dried animal dung may also be to provide shelter; a parachute canopy if available; compass, signal mirror, used as fuel. and signal devices; map; pencil; knife; salt; sun-glasses; first aid kit; a watch; and a flashlight.

Clothing

During the desert summer you'll travel only at night and rest during the day. Don't hurry, follow the easiest route, and avoid soft sand areas Don't discard any of your clothing. You require it for protection and rough terrain as much as possible. Determine your position accurately against sunburn, heat, sand, and insects. Keep your head, neck, and body and mark it on the map before setting out. Decide on a course and stick covered during the day, and you'll last longer on less water. Wear loose, to it. Your compass and the stars will be your greatest navigational aids. light-coloured clothing if possible, using parts of parachutes if available. When deciding your objective, it is best to choose one that is easy to find, Your T-shirt makes a good neck drape. If you have no hat make one such as a coat or road which can be followed until habitation is reached, from a parachute or any other material available, similar to the type worn rather than a specific point like an oasis which may be very difficult to by the Arabs. You can adapt a pilot chute for use as a parasol. find or pinpoint. Eyes must be protected from both direct and reflected glare. Sunglasses must be worn throughout the day. If they are not available a piece of Natives cloth with slits cut for the eyes (just large enough to allow a penny to pass through) makes a good eyeshield. Smearing soot under the eyes helps Another important phase of desert survival is obtaining assistance to reduce the glare. from others. If you can contact any of the natives who live on the desert, Your feet should be kept in good shape. Rest your feet often. Remove your chances of survival will greatly increase. Generally their sense of shoes, dry out socks and turn them inside out, and try to keep the inside hospitality is very strong, and they will provide you with water, food, and shelter. Their customs and religion are much different from ours, so try of shoes free from sand. If your shoes wear out, or you lose them, not to offend them in any way. improvise a type of sandal from aircraft rubber flooring, parachute material, Desert survival is difficult but by no means impossible. However, or any other available material. you must be ready to solve the problems of water and exposure as well If you are caught in a dust storm, cover your mouth and nose. Button up your clothing tightly, and lie down back to the wind. To avoid sand as the other desert hazards. from drifting around you, roll about from time to time. Owing to the extremes in temperatures, the need for additional clothing

DAILY WATER REQUIREMENT TO
will be felt at night.
MAINTAIN WATER BALANCE

		Mean Temperature		Litres per			
Food		Degrees C		24 Hours			
				First check			
Food	on the desert is always a problem.						
	in,						
any	ration packs or fresh food.	Any	open	foods	or opened		
	should	35		5			
Big							
be eaten first as they will not keep long.							
32 3.7							
Although birds and animals are scarce							
vegetation,	they may sometimes	be found.		Small	with		
				in areas	burrowing		
water							
little							
animal							

or
come

29

2.6

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out at dawn or dusk and can be caught in snares.

. 27 1.4
Edible plant food is rare in the desert. Grasses are edible, and some

24 11
areas have edible wild tulips or onions. Avoid plants with milky or
coloured saps as they are likely to be poisonous. Remember the rules with
in the desert the mean temperature can be taken as 10°C below the
regard to eating if water is scarce.
daily maximum.

In hot deserts you need a minimum of 4.5 2 of water a day. If you

Travel

walk in the cool desert night you can get approximately 32 km on that
The decision to travel or remain with the aircraft will in all likeli-
daily amount of water. If you travel in day-time heat, you'll be lucky to.
hood be difficult to make. Travel should be undertaken only if it is certain
get 16-km on that same amount of water.

' |

3

TROPICAL
SURVIVAL

TROPICAL

SURVIVAL

Introduction

Uninhabited tropical areas are not as difficult to survive in as many people believe. Some people visualize the tropics as an enormous and forbidding jungle through which each step taken must be hacked out. Actually, much of the tropics is not jungle. It is true that travel in jungle areas is very difficult, but normally there is little danger from animals. Tropical terrain varies from forest jungle, mangrove, or other swamps to Open grassy plains or semi-arid bushland.

The real dangers of the tropics are not the animals or snakes, which most people have heard wild tales about, but rather the insects. Many of the insects carry infection and disease. One of the worst diseases is malaria transmitted by the mosquito. Many people are frightened by the howls, screams, and other sounds made by birds, animals, and insects. However, this is only psychological and usually no real danger exists.

Immediate Action

The first decision to be made after a crash landing will probably be whether to travel or stay with the aircraft. If a landing is made in an area where neither the aircraft nor signals can be easily sighted from the air and if there are no injuries, then probably it would be wise to travel.

Injuries must be taken care of first. In tropical areas even the slightest wratch can cause a serious infection within hours, so first aid is the first requirement after landing. Don't leave the crash area without carefully blazing or marking a route so that you can return to the crash site or others can follow your trail.

Shelter

Night in the jungle comes very quickly, so prepare for bed early. In the jungle more rest and sleep are required to keep up energy and strength in order to maintain resistance to disease.

Try to locate a camp site on a knoll or high spot in a clearing well away from the swamps. There will be fewer mosquitoes and other insects, and the ground will be dryer. Such an area will be necessary for ground tagnals too.

Do not plan to sleep on the ground. Construct a bed by covering a Good tinder may be found in the fibres at the bases of palm leaves, pile of brush with layers of palm fronds or other broad leaves. A better The inside of termite nests also makes good kindling. Keep a good supply bed can be constructed by making a frame of poles and covering the top of wood handy, and keep it dry by stowing it in your shelter or beneath with long, spineless, palm leaves, to a depth of four or five layers. Cut broad green leaves. Dry out wet kindling and fuel near your fire. the corner upright poles Jong enough to support a mosquito net or parachute covering. A hammock provides a good type of bed and can be made

Food and Water

from a parachute strung between two trees.

Normally both food and water are plentiful in the tropics. Water is A good rainproof shelter can be built by covering an A-type frame-available from the numerous streams, springs, lakes, pools, and swamps, work with a good thickness of palm or other broad leaf, pieces of bark, but it is not safe to drink until it has been purified. Some water may be or mats of grass. This is similar to a brush type lean-to made from spruce discoloured or turbid and may be partially cleared by filtering through a boughs, only it has two sides instead of one.

improvised filter such as parachute cloth. One may also obtain water

If you stay with your aircraft, it may be used for a shelter and will from some plants which can be used without further treatment. Coconuts probably provide a dry area in the wet jungle forest. Try to make it contain water, and the green unripe coconuts about the size of a grapefruit mosquito proof by covering the opening with netting or parachute material. are the best. In addition vines are often a good source of water; however Do not camp too near a stream or pond, especially during the rainy never drink from a vine that has a milky sap. Water may also be collected season. Do not build a shelter under dead trees or with dead limbs, nor during a rain by digging a hole and lining it with a tarpaulin or a piece under a coconut tree. A falling coconut can disable a person. of canvas.

Clothing protects against exposure, insects, and pests. Keep pant Animal trails often lead to water. Food is usually easy to procure .

cuffs tucked into the tops of the stockings, and tie them securely or in the jungle. There are usually a number of different types of fruits and improvise puttees to keep out ticks and leeches. Keep sleeves rolled down vegetables. Some of the common edible ones are: sago palm or palm and buttoned. This also helps to protect against scratches from thorns, cabbage; bananas; bamboo shoots; coconuts; and papaya. Any food > brambles, etc.

monkey eats can be eaten by man.

When clothes are removed, make a thorough inspection of the body

There are edible fish in most of the jungle streams. You need have for ticks, chiggers, insects, leeches, or any other vermin. Check the clothing no fear about poisonous fish because they are found only in salt water. too, and remove any insects that may have got on it.

Try to have the clothing dry before nightfall to avoid discomfort

Travel

from the cold. It is most important to keep clothing clean, dry, and in Travel in the tropics can be difficult because of heavy undergrowth, good repair. Dirty clothes not only rot quickly but may lead to skin disease. heat and humidity, swamps, mangroves, and the lack of landmarks.

The most useful aids to travel are: a machete to help cut a route:

Signals

find food, and make a raft; a compass; a first aid kit in case of fever or As in most other survival situations the aircraft radio, if working, is infection; good foot-gear; and a hammock or material to construct one. the best signal device. Emergency radios are not very efficient in jungle

If possible follow the downward flow of a stream as it likely flows areas. Pyros should be kept handy and dry. A signal area may be very towards larger bodies of water and consequently towards settlements. When difficult to find. However, parachutes or contrasting coloured objects such possible, construct a raft from bamboo or light wood and float down the as dinghies stretched across streams or bays or placed in ponds may attract stream, as this js the fastest and easiest method of transportation in tropical attention. Smoke or fire signals in clearings are also effective. Make use forest areas.

of every Signal device that is available.

Avoid such obstacles as thickets, mangroves, and swamps. Stop early and prepare a night camp as it gets dark early and quickly in the tropics.

Fires

A fire in the jungle is most desirable. It provides warmth during chilly

¥

nights, serves for cooking, and helps keep away mosquitoes and curiou animals. A big roaring fire is not necessary, because a small one serves the same purpose and is easier to maintain.

Fuel is usually plentiful. The fire lighting problem may be difficult during the rainy season when dry fuel is hard to locate. Many of the large trees whether dead or alive have hollow trunks. Cut strips of the dry inner lining for tinder. When the fire is going well, wet wood can be added. Dry wood may be found hanging in the network of vines or lying

on bushes.

Don't use bamboo for fuel. It burns too quickly, emits dangerous fumes, and may explode,

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KNOTS AND
SPLICES

CHAPTER 15
KNOTS AND SPLICES

Knot Requirements

The four basic requirements for knots are:

- (a) they must be easy to tie or untie;
- (b) they can be tied in the middle of a length of rope;
- (c) they can be tied when the rope is under tension; and
- (d) they can be tied in such a fashion that the rope will not cut itself when under strain.

Classes of Knots

There are five basic classes of knots. They are used for:

- (a) joining two ends of rope of the same diameter (reef);
- (b) joining two ends of rope of different diameters (bends);
- (c) forming running loops (bowline group);
- (d) forming stationary loops (bowline group); and
- (e) securing one end of the rope to a point (hitches).

8

8 a

a Thumb Knot,
Stops unlaying of the
Tope.

b Figure of Eight.
Prevents the rope
from being pulled
through the pulley.

c Reef Knot.
For joining two
ropes of equal
diameter.

d@ Sheet Bend. For joining two ropes of unequal diameters.	¢ Hawser Bend. For tying two large ropes together.	f£ Bowline. For making a non- slip loop in the end of a rope.	
id Turn—Two = Reinebes. Mooring knot.	Hi wo Half		j Timber Hitch. For hauling or towing timber.
t Sheep Shank. For shortening a rope that is tied at both ends. g Running Bowline. For making a running loop in the end of a rope.		h Clove Hitch. Mooring knot.	

Splices

Back Splice

20 cm;

(a) unlay the rope for approximately 20 cm at the end;

where the unlaying stops by putting the

(b) make the crown knot;

between the strands of the other end

(c) splice the ends of the rope into the main rope, taking them over one and under the next one;

under one with all six strands;

(d) when all three strands have been spliced in once, this is known as

more times;

a tuck;

do two half tucks with the half

(e) carry out this procedure for four full tucks;

(ff) then split the strands in half and taking one half of each strand

a centimetre; and

do two more tucks—these are called half tacks, because you use

only half the rope, which tapers off the end of the splice;

starting two centimetres above the

do this by starting a centimetre above the cut ends and pulling

ending a centimetre below the splice,

(8)

the cord as tight as possible and keeping the tums close together while working away from the end of the rope to a centimetre past the end of the splice.

Back Splice

Short Splice

(a) unlay both ends of the rope for

(b) marry the ends of the rope

strands of one end of the rope

of the rope;

(c) splice by going over one and

(d) carry out this procedure three

(e) divide each strand in half and

strands;

(f) cut off all strands to within

(g) serve the splice with cord,

start of the splice and